STAT 460: Statistical Inference I

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Contents

1 Introduction and Review of Basic Probability **1**
1.1 Sample Spaces and Events . . . 1
1.2 Axioms of Probability . . . 1
1.3 Independence . . . 3
1.4 Random Variables . . . 3
1.5 Multivariate Distributions and Independence . . . 3
1.6 Conditional Distributions . . . 4
1.7 Transformations . . . 4
1.8 The Multivariate Normal Distribution . . . 4

2 Expectation and Inequalities **5**
2.1 Expectation . . . 5
2.2 Moments and Moment Generating Functions . . . 5
2.3 Probability Inequalities . . . 6
2.4 Confidence Intervals from Concentration Bounds . . . 7

3 Convergence of Random Variables **8**
3.1 Modes of Convergence . . . 8
3.2 Slutsky's Lemma . . . 9
3.3 Stochastic o and O Symbols . . . 9
3.4 Characteristic Functions . . . 10
3.5 Central Limit Theorem . . . 10

4 The Delta Method and the Big Picture **11**
4.1 The Delta Method . . . 11
4.2 Statistical Models and Estimation . . . 12
4.3 Confidence Sets . . . 12

5 Nonparametric Inference **13**
5.1 Estimating the CDF and Functionals of the CDF . . . 13
5.2 Bootstrap Methods . . . 15
5.3 Consistency of Bootstrap Confidence Intervals . . . 16

5.4 Bootstrap Validity for the Sample Mean 16
5.5 Lindeberg–Feller CLT 17

6 Parametric Inference **18**
6.1 Parametric Frequentist Inference 18
6.2 Parametric Bayesian Inference 20
6.3 Statistical Decision Theory 21

7 Prediction and Model Selection **23**
7.1 Empirical Risk Minimization and Cross-Validation 23
7.2 Nonparametric Curve Estimation 25
7.3 Nonparametric Regression: Linear Smoothers and Basis Functions 27
7.4 Shrinkage and Regularization 28
7.5 The James–Stein Estimator, Shrinkage, and Ridge Regression 28
7.6 Stein's Lemma, SURE, and James–Stein Analysis 30

8 General Asymptotic Theory **32**
8.1 Consistency of M- and Z-estimators 32
8.2 Asymptotic Normality of M- and Z-estimators 34
8.3 Step Estimators and Examples 35
8.4 Asymptotic Normality of Z-estimators with Nuisance Parameters 37

9 Special Topics in Likelihood **41**
9.1 Time-to-Event Data and Survival Analysis 41
9.2 Missing Data, Latent Variables, and the EM Algorithm 43

Chapter 1

Introduction and Review of Basic Probability

1.1 Sample Spaces and Events

Definition (Probability model). An **experiment** is modeled by a **sample space** Ω of outcomes $\omega \in \Omega$, together with a collection of **events** $A \subseteq \Omega$.

Given events $A, B \subseteq \Omega$:

- **Complement:** $A^c = \Omega \setminus A$
- **Union:** $A \cup B = \{\omega : \omega \in A \text{ or } \omega \in B\}$
- **Intersection:** $A \cap B = \{\omega : \omega \in A \text{ and } \omega \in B\}$
- **Set difference:** $A \setminus B = A \cap B^c$

A sequence $A_1, A_2, \ldots$ is **disjoint** if $A_i \cap A_j = \emptyset$ for $i \neq j$, and a **partition** if also $\bigcup_{i \geq 1} A_i = \Omega$.

1.2 Axioms of Probability

Definition (Probability measure). A **probability measure** P assigns to each event $A \subseteq \Omega$ a real number $P(A)$ satisfying Kolmogorov's axioms:

1. $P(A) \geq 0$ for every event A.
2. $P(\Omega) = 1$.
3. **Countable additivity:** If $A_1, A_2, \ldots$ are disjoint, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Proposition (Properties of P). Let P be a probability measure. Then:

1. $P(\emptyset) = 0$.
2. **Finite additivity:** $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$.
3. $P(A^c) = 1 - P(A)$.
4. **Monotonicity:** $A \subseteq B \Rightarrow P(A) \leq P(B)$.
5. **Inclusion-exclusion:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. (1) Take $A_1 = \Omega$ and $A_i = \emptyset$ for $i \geq 2$. These are disjoint and $\bigcup_i A_i = \Omega$, so by countable additivity, $P(\Omega) = P(\Omega) + \sum_{i \geq 2} P(\emptyset)$. Since $P(\Omega) = 1 < \infty$, we must have $P(\emptyset) = 0$.
(2) Apply countable additivity to $A, B, \emptyset, \emptyset, \ldots$
(3) Since A and A^c are disjoint and $A \cup A^c = \Omega$, finite additivity gives $P(A) + P(A^c) = 1$.
(4) Write $B = A \cup (B \setminus A)$, a disjoint union. By finite additivity, $P(B) = P(A) + P(B \setminus A) \geq P(A)$.
(5) Write $A \cup B = A \cup (B \setminus A)$ disjointly, so $P(A \cup B) = P(A) + P(B \setminus A)$. Since $B = (A \cap B) \cup (B \setminus A)$ disjointly, $P(B \setminus A) = P(B) - P(A \cap B)$. Substituting gives the result. □

Definition (Monotone sequences). A sequence of sets is **monotone increasing** if $A_1 \subseteq A_2 \subseteq \cdots$, with limit $\lim_n A_n = \bigcup_{i \geq 1} A_i$. It is **monotone decreasing** if $A_1 \supseteq A_2 \supseteq \cdots$, with limit $\lim_n A_n = \bigcap_{i \geq 1} A_i$.

Theorem (Continuity of probability). If $A_1, A_2, \ldots$ is monotone increasing or monotone decreasing, then

$$\lim_{n \to \infty} P(A_n) = P\left(\lim_{n \to \infty} A_n\right).$$

Proof. **Increasing case.** Define $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Then the B_k are disjoint, and $\bigcup_{k=1}^{n} B_k = A_n$, $\bigcup_{k=1}^{\infty} B_k = \bigcup_{k=1}^{\infty} A_k$. By countable additivity,

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(B_k) = \lim_{n \to \infty} \sum_{k=1}^{n} P(B_k) = \lim_{n \to \infty} P(A_n).$$

Decreasing case. Apply the increasing case to $A_1^c \subseteq A_2^c \subseteq \cdots$, using $P(A_n^c) = 1 - P(A_n)$ and $\bigcup A_k^c = (\bigcap A_k)^c$. □

Definition (Uniform distribution on finite Ω). When Ω is finite, the **uniform distribution** is $P(A) = |A|/|\Omega|$ for $A \subseteq \Omega$.

1.3 Independence

Definition (Independence). A collection of events $\{A_i : i \in I\}$ is **independent** if

$$P\left(\bigcap_{i\in I} A_i\right) = \prod_{i\in I} P(A_i).$$

1.4 Random Variables

Definition (Random variable). A **random variable** is a measurable function $X : \Omega \to \mathbb{R}$. The **distribution** of X is determined by $P(X \in A) = P(X^{-1}(A))$ for $A \subseteq \mathbb{R}$.

Definition (CDF, PMF, PDF). The **cumulative distribution function (CDF)** of X is $F_X(x) = P(X \leq x)$.

- X is **discrete** if it takes countably many values; the **PMF** is $f_X(x) = P(X = x)$, and $F_X(x) = \sum_{x_i \leq x} f_X(x_i)$.
- X is **continuous** if there exists $f_X \geq 0$ with $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$ and $P(a < X < b) = \int_a^b f_X(x)\,\mathrm{d}x$. The function f_X is the **PDF**, and $F_X(x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$.

Definition (Quantile function). The **quantile function** (inverse CDF) is

$$F^{-1}(q) = \inf\{x : F(x) > q\}.$$

When F is continuous and strictly increasing this is the functional inverse of F.

1.5 Multivariate Distributions and Independence

Definition (Joint and marginal distributions). For a random vector (X, Y) with joint PDF $f_{X,Y}$, the **marginal PDF** of X is $f_X(x) = \int f_{X,Y}(x, y)\,\mathrm{d}y$.

Definition (Independence of random variables). X and Y are **independent**, written $X \perp Y$, if $P(X \in A, Y \in B) = P(X \in A)\,P(Y \in B)$ for all $A, B \subseteq \mathbb{R}$.

Theorem. $X \perp Y$ if and only if $f_{X,Y}(x, y) = f_X(x)\,f_Y(y)$ for all $(x, y) \in \mathbb{R}^2$.

Definition (i.i.d.). $X_1, \ldots, X_n$ are **independent and identically distributed (i.i.d.)** if they are mutually independent and share the same marginal CDF F.

1.6 Conditional Distributions

Definition (Conditional PDF/PMF). Given joint PDF/PMF $f_{X,Y}$ with marginal $f_Y(y) > 0$, the **conditional PDF/PMF** of X given $Y = y$ is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

1.7 Transformations

Given X with PDF f_X and $Y = r(X)$, the distribution of Y is found by:

1. For each y, find $A_y = \{x : r(x) \leq y\}$.
2. Compute $F_Y(y) = \displaystyle\int_{A_y} f_X(x)\,\mathrm{d}x$.
3. Set $f_Y(y) = F_Y'(y)$.

When r is strictly monotone with inverse $s = r^{-1}$,

$$f_Y(y) = f_X(s(y)) \left|\frac{\mathrm{d}s}{\mathrm{d}y}\right|.$$

1.8 The Multivariate Normal Distribution

Definition (Multivariate normal). $X \in \mathbb{R}^k$ has a **multivariate normal distribution** $X \sim \mathcal{N}(\mu, \Sigma)$ with mean $\mu \in \mathbb{R}^k$ and symmetric positive definite covariance $\Sigma \in \mathbb{R}^{k\times k}$ if

$$f_X(x; \mu, \Sigma) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\left(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right).$$

Theorem (Affine transformations of normals). Let $X \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^k$, $c \in \mathbb{R}^m$, and $B \in \mathbb{R}^{m\times k}$. Then

$$Y = c + BX \sim \mathcal{N}(c + B\mu,\ B\Sigma B^\top).$$

Corollary. If $X \sim \mathcal{N}(\mu, \Sigma)$ then $Z = \Sigma^{-1/2}(X - \mu) \sim \mathcal{N}(0, I)$, where $\Sigma^{1/2}$ is the symmetric square root from the eigendecomposition $\Sigma = U\Lambda U^\top$.

Proof. Apply Theorem 1.8 with $c = -\Sigma^{-1/2}\mu$ and $B = \Sigma^{-1/2}$. Then $B\mu + c = 0$ and $B\Sigma B^\top = \Sigma^{-1/2}\Sigma(\Sigma^{-1/2})^\top = \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I$. $\square$

Chapter 2

Expectation and Inequalities

2.1 Expectation

Definition (Expectation). The **expectation** of a random variable X with CDF F_X is

$$E(X) = \int x \, \mathrm{d}F_X(x) = \begin{cases} \sum_x x \, f_X(x) & \text{if } X \text{ is discrete,} \\ \int x \, f_X(x) \, \mathrm{d}x & \text{if } X \text{ is continuous,} \end{cases}$$

provided $\int |x| \, \mathrm{d}F_X(x) < \infty$.

For $Y = r(X)$: $E(Y) = \int r(x) \, \mathrm{d}F_X(x)$.

2.2 Moments and Moment Generating Functions

Definition (MGF). The **moment generating function (MGF)** of X is

$$\psi_X(t) = E(e^{tX}), \quad t \in \mathbb{R},$$

defined when finite in an open interval around $t = 0$. When it exists, $\psi_X^{(k)}(0) = E(X^k)$.

Proposition (MGF of a linear combination). Let $X_1, \ldots, X_n$ be independent with MGFs ψ_{X_i}. For constants $a_i, b_i \in \mathbb{R}$, let $Y = \sum_{i=1}^n a_i X_i + b_i$. Then

$$\psi_Y(t) = \prod_{i=1}^n e^{b_i t} \, \psi_{X_i}(a_i t).$$

Proof. By independence, $E(e^{tY}) = E\left(\prod_i e^{t(a_i X_i + b_i)}\right) = \prod_i e^{b_i t} E(e^{a_i t X_i}) = \prod_i e^{b_i t} \psi_{X_i}(a_i t)$. $\square$

Example (Gamma MGF and closure under summation). The Gamma(α, β) distribution with PDF $f_X(x) = \dfrac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$ has MGF

$$\psi_X(t) = (1 - \beta t)^{-\alpha}, \quad t < 1/\beta.$$

If $X_i \sim \text{Gamma}(\alpha_i, \beta)$ are independent, then $Y = \sum_i X_i \sim \text{Gamma}(\sum_i \alpha_i, \beta)$, since $\psi_Y(t) = \prod_i (1 - \beta t)^{-\alpha_i} = (1 - \beta t)^{-\sum_i \alpha_i}$.

2.3 Probability Inequalities

Theorem (Markov's inequality). If $X \geq 0$ and $E(X) < \infty$, then for any $t > 0$,

$$P(X > t) \leq \frac{E(X)}{t}.$$

Proof. $E(X) = \int_0^\infty x \, \mathrm{d}F(x) \geq \int_t^\infty x \, \mathrm{d}F(x) \geq t \int_t^\infty \mathrm{d}F(x) = t \, P(X > t).$ □

More generally, if $f : \mathbb{R} \to \mathbb{R}_+$ is increasing and $E(f(X))$ exists, then

$$P(X > t) \leq \frac{E(f(X))}{f(t)}.$$

Corollary (Chebyshev's inequality). For any $t > 0$,

$$P(|X - E(X)| > t) \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Apply Markov's inequality to $(X - E(X))^2$ with threshold t^2. □

Definition (Chernoff bound). The **Chernoff bound** applies Markov's inequality to e^{aX}: for any $a > 0$,

$$P(X > t) \leq e^{-at} \psi_X(a), \qquad P(X > t) \leq \inf_{a>0} e^{-at} \psi_X(a).$$

For the lower tail, take $a < 0$: $P(X < t) \leq \inf_{a<0} e^{-at} \psi_X(a)$.

Theorem (Hoeffding's inequality). Let $X_1, \ldots, X_n$ be independent with $E(X_i) = 0$ and $P(a_i \leq X_i \leq b_i) = 1$. For any $t > 0$,

$$P\left(\sum_{i=1}^n X_i > t\right) \leq \exp\left(-at + \frac{a^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right)$$

for any $a > 0$. Optimizing over a:

$$P\left(\sum_{i=1}^n X_i > t\right) \leq \exp\left(\frac{-2t^2}{\sum_i (b_i - a_i)^2}\right).$$

Proof. Start from the Chernoff bound. The key step is to bound the MGF of a bounded mean-zero random variable: if $E(X) = 0$ and $a \leq X \leq b$, then $E(e^{aX}) \leq e^{a^2(b-a)^2/8}$ (using $1 + x \leq e^x$ and convexity). Applying this to each X_i and multiplying gives the result. Optimizing over a yields the stated bound. □

Corollary (Hoeffding for Bernoulli means). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bern}(p)$. For any $t > 0$,

$$P(|\overline{X}_n - p| > t) \leq 2e^{-2nt^2}.$$

Proof. Apply Theorem 2.3 to $Y_i = (X_i - p)/n$ (so $E(Y_i) = 0$ and $b_i - a_i = 1/n$), giving $P(\overline{X}_n - p > t) \leq e^{-2nt^2}$. The same bound holds for the lower tail, and a union bound gives the two-sided result. □

2.4 Confidence Intervals from Concentration Bounds

Inverting a two-sided bound directly yields a confidence interval. For the Hoeffding bound above, setting

$$t_n = \sqrt{\frac{1}{2n} \log\left(\frac{2}{\alpha}\right)}$$

gives $P(|\overline{X}_n - p| > t_n) \leq \alpha$, so $(\overline{X}_n - t_n, \overline{X}_n + t_n)$ is a valid $1 - \alpha$ confidence interval for p.

Chapter 3

Convergence of Random Variables

3.1 Modes of Convergence

Let $X_1, X_2, \ldots$ and X be random vectors in $\mathbb{R}^k$.

Definition (Convergence in distribution). $X_n \xrightarrow{d} X$ if $\lim_{n\to\infty} P(X_n \le t) = P(X \le t)$ at every $t \in \mathbb{R}^k$ where $t \mapsto P(X \le t)$ is continuous.

Definition (Convergence in probability). $X_n \xrightarrow{p} X$ if for every $\varepsilon > 0$, $\lim_{n\to\infty} P(\|X_n - X\| > \varepsilon) = 0$.

Definition (Convergence in quadratic mean (L^2)). $X_n \xrightarrow{qm} X$ if $\lim_{n\to\infty} E[\|X_n - X\|^2] = 0$.

Definition (Almost sure convergence). $X_n \xrightarrow{a.s.} X$ if $P\{\omega : \liminf_n X_n(\omega) = \limsup_n X_n(\omega) = X(\omega)\} = 1$.

Theorem (Relations between modes of convergence). Let X_n, X, Y be random vectors and c a constant.

1. $X_n \xrightarrow{a.s.} X \;\Rightarrow\; X_n \xrightarrow{p} X$.
2. $X_n \xrightarrow{qm} X \;\Rightarrow\; X_n \xrightarrow{p} X$.
3. $X_n \xrightarrow{p} X \;\Rightarrow\; X_n \xrightarrow{d} X$.
4. $X_n \xrightarrow{p} c \;\Leftrightarrow\; X_n \xrightarrow{d} c$.
5. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $(X_n, Y_n) \xrightarrow{d} (X, c)$.
6. If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $(X_n, Y_n) \xrightarrow{p} (X, Y)$.

Proof of (2). By Chebyshev's inequality, $P(\|X_n - X\| > \varepsilon) \le E[\|X_n - X\|^2]/\varepsilon^2 \to 0$. $\square$

Proof of (3). For any continuity point t of $x \mapsto P(X \le x)$ and $\delta > 0$,

$$P(X_n \le t) \le P(X \le t + \delta) + P(\|X_n - X\| > \delta).$$

Similarly $P(X_n \le t) \ge P(X \le t - \delta) - P(\|X_n - X\| > \delta)$. As $n \to \infty$, the probability terms vanish; then let $\delta \to 0$ using continuity of $P(X \le \cdot)$ at t. □

Theorem (Weak Law of Large Numbers). Let $X_1, \ldots, X_n$ be i.i.d. with finite mean $E(X)$ and variance. Then $\overline{X}_n \xrightarrow{p} E(X)$.

Proof. $E[(\overline{X}_n - E(X))^2] = \mathrm{Var}(\overline{X}_n) = \mathrm{Var}(X)/n \to 0$, so $\overline{X}_n \xrightarrow{qm} E(X)$, and (2) of Theorem 3.1 gives convergence in probability. □

Theorem (Continuous mapping theorem). Let $g : \mathbb{R}^k \to \mathbb{R}^m$ be continuous on a set C with $P(X \in C) = 1$. Then:

1. $X_n \xrightarrow{d} X \;\Rightarrow\; g(X_n) \xrightarrow{d} g(X)$.
2. $X_n \xrightarrow{p} X \;\Rightarrow\; g(X_n) \xrightarrow{p} g(X)$.
3. $X_n \xrightarrow{a.s.} X \;\Rightarrow\; g(X_n) \xrightarrow{a.s.} g(X)$.

3.2 Slutsky's Lemma

Lemma (Slutsky). Let X_n, X, Y_n be random vectors and c a constant. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then:

1. $X_n + Y_n \xrightarrow{d} X + c$.
2. $Y_n X_n \xrightarrow{d} cX$.
3. $X_n / Y_n \xrightarrow{d} X/c$, provided $c \neq 0$.

Proof. Since $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, by Theorem 3.2(5) we have $(X_n, Y_n) \xrightarrow{d} (X, c)$. Each claim then follows from the continuous mapping theorem applied to the respective continuous function $(x, y) \mapsto x + y$, $(x, y) \mapsto xy$, $(x, y) \mapsto x/y$. □

3.3 Stochastic o and O Symbols

Definition (Bounded in probability, O_P). A sequence X_n is **bounded in probability** (or **uniformly tight**), written $X_n = O_P(1)$, if for each $\varepsilon > 0$ there exists $M_\varepsilon < \infty$ such that $\sup_n P(\|X_n\| > M_\varepsilon) < \varepsilon$.
More generally, $X_n = O_P(R_n)$ if $X_n / R_n = O_P(1)$.

Definition (Convergence in probability, o_P). $X_n = o_P(1)$ if $X_n \xrightarrow{p} 0$. More generally, $X_n = o_P(R_n)$ if $X_n / R_n \xrightarrow{p} 0$.

Proposition (o_P/O_P calculus). The following rules hold (where = means "has the same asymptotic order as at most"):

$$o_P(1) + o_P(1) = o_P(1), \quad o_P(1) \cdot O_P(1) = o_P(1), \quad o_P(R_n) = R_n \cdot o_P(1).$$

If $R_n(X_n - \theta) = O_P(1)$ for $R_n \to \infty$, then $X_n \xrightarrow{p} \theta$.

3.4 Characteristic Functions

Definition (Characteristic function). The **characteristic function** of $X \in \mathbb{R}^k$ is

$$\varphi_X(t) = E[e^{it^\top X}], \quad t \in \mathbb{R}^k,$$

where $i = \sqrt{-1}$.

Key properties:

1. φ_X uniquely determines the distribution of X.
2. $X_n \xrightarrow{d} X$ if and only if $\varphi_{X_n}(t) \to \varphi_X(t)$ for all t.
3. If $\varphi_{X_n}(t) \to \varphi(t)$ pointwise and φ is continuous at 0, then φ is the characteristic function of some X and $X_n \xrightarrow{d} X$. (**Lévy's continuity theorem.**)
4. The distribution is determined by $\{t^\top X : t \in \mathbb{R}^k\}$. (**Cramér–Wold device.**)

Example (Normal characteristic function). If $X \sim \mathcal{N}_k(\mu, \Sigma)$, then

$$\varphi_X(t) = \exp\left(it^\top \mu - \tfrac{1}{2} t^\top \Sigma t\right).$$

3.5 Central Limit Theorem

Theorem (Multivariate CLT). Let $X_1, X_2, \ldots$ be i.i.d. random vectors in $\mathbb{R}^k$ with mean $E(X)$ and covariance matrix Σ. Then

$$\sqrt{n}(\overline{X}_n - E(X)) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Proof sketch, $k = 1$. Let $\mu = E(X)$, $\sigma^2 = \mathrm{Var}(X)$. The characteristic function of $\sqrt{n}(\overline{X}_n - \mu)$ is

$$\varphi_{\sqrt{n}(\overline{X}_n - \mu)}(t) = \left[\varphi_{X-\mu}\left(\tfrac{t}{\sqrt{n}}\right)\right]^n.$$

Taylor-expand $\varphi_{X-\mu}(s)$ around $s = 0$: since $E(X - \mu) = 0$, we get $\varphi_{X-\mu}(s) = 1 - \dfrac{\sigma^2 s^2}{2} + o(s^2)$. Substituting $s = t/\sqrt{n}$:

$$\varphi_{\sqrt{n}(\overline{X}_n - \mu)}(t) = \left(1 - \frac{\sigma^2 t^2}{2n} + o(1/n)\right)^n \to e^{-\sigma^2 t^2/2},$$

which is the characteristic function of $\mathcal{N}(0, \sigma^2)$. The multivariate case follows via the Cramér–Wold device. $\square$

Chapter 4

The Delta Method and the Big Picture

4.1 The Delta Method

Definition (Differentiability). A function $\varphi : \mathbb{R}^k \to \mathbb{R}^m$ is **differentiable at** θ if there exists a linear map $\varphi'_\theta : \mathbb{R}^k \to \mathbb{R}^m$ (given by the Jacobian matrix) such that

$$\varphi(\theta + h) - \varphi(\theta) = \varphi'_\theta(h) + o(\|h\|), \quad h \to 0.$$

Theorem (Delta method). Let $\varphi : \mathbb{R}^k \to \mathbb{R}^m$ be differentiable at θ, and let T_n be random vectors in the domain of φ. If $r_n(T_n - \theta) \xrightarrow{d} T$ for $r_n \to \infty$, then:

1. $r_n(\varphi(T_n) - \varphi(\theta)) \xrightarrow{d} \varphi'_\theta(T)$.
2. $r_n(\varphi(T_n) - \varphi(\theta)) - \varphi'_\theta(r_n(T_n - \theta)) \xrightarrow{p} 0$.

Proof. Since $r_n(T_n - \theta) \xrightarrow{d} T$, it is $O_P(1)$ by Prohorov's theorem. Hence $T_n - \theta \xrightarrow{p} 0$ (since $r_n \to \infty$). Define the remainder $R(h) = \varphi(\theta + h) - \varphi(\theta) - \varphi'_\theta(h)$. By differentiability, $R(h) = o(\|h\|)$ as $h \to 0$. Since $T_n - \theta \xrightarrow{p} 0$, we get $R(T_n - \theta) = o_P(\|T_n - \theta\|)$. Multiplying by r_n:

$$r_n(\varphi(T_n) - \varphi(\theta)) - \varphi'_\theta(r_n(T_n - \theta)) = r_n\, R(T_n - \theta) = r_n \|T_n - \theta\| \cdot o_P(1) = O_P(1) \cdot o_P(1) = o_P(1).$$

This proves (ii). For (i), note $\varphi'_\theta(r_n(T_n - \theta)) \xrightarrow{d} \varphi'_\theta(T)$ by the continuous mapping theorem (matrix multiplication is continuous), and then Slutsky's lemma applied to (ii) gives the result. $\square$

Corollary (Normal delta method). If $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}_k(\mu, \Sigma)$ and $\varphi : \mathbb{R}^k \to \mathbb{R}^m$ is differentiable at θ, then

$$\sqrt{n}(\varphi(T_n) - \varphi(\theta)) \xrightarrow{d} \mathcal{N}_m\big(\varphi'_\theta \mu,\ \varphi'_\theta \Sigma (\varphi'_\theta)^\top\big)\,.$$

4.2 Statistical Models and Estimation

Definition (Statistical model). A **statistical model** $\mathcal{F}$ is a set of probability distributions. A **parametric model** is one indexed by a finite-dimensional parameter:

$$\mathcal{F} = \left\{P_\theta : \theta \in \Theta \subseteq \mathbb{R}^k\right\}.$$

A **nonparametric model** cannot be so parameterized (e.g., $\mathcal{F} = \{\text{all CDFs}\}$). A **semiparametric model** has both a parametric component of interest and a nonparametric nuisance component.

Definition (Estimator, bias, consistency). An **estimator** $\hat{\theta}_n = g(X_1, \ldots, X_n)$ is a function of the data. Its **bias** is

$$\text{bias}(\hat{\theta}_n) = E_\theta(\hat{\theta}_n) - \theta.$$

$\hat{\theta}_n$ is **unbiased** if its bias is zero; **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta$.

Definition (Standard error and MSE). The **standard error** is $\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}_\theta(\hat{\theta}_n)}$. The **mean squared error** is

$$\text{mse}(\hat{\theta}_n) = E_\theta[(\hat{\theta}_n - \theta)^2] = \text{bias}^2(\hat{\theta}_n) + \text{Var}_\theta(\hat{\theta}_n).$$

Theorem. If $\text{mse}(\hat{\theta}_n) \to 0$ as $n \to \infty$, then $\hat{\theta}_n$ is consistent.

Proof. By Chebyshev, $P(|\hat{\theta}_n - \theta| > \varepsilon) \leq \text{mse}(\hat{\theta}_n)/\varepsilon^2 \to 0$. □

Definition (Asymptotic normality). $\hat{\theta}_n$ is **asymptotically normal** if

$$\frac{\hat{\theta}_n - \theta}{\text{se}(\hat{\theta}_n)} \xrightarrow{d} \mathcal{N}(0, 1).$$

4.3 Confidence Sets

Definition ($1-\alpha$ confidence interval). A random interval $C_n = (a(X_1, \ldots, X_n),\ b(X_1, \ldots, X_n))$ is a $1 - \alpha$ **confidence interval** if $P_\theta(\theta \in C_n) \geq 1 - \alpha$ for all $\theta \in \Theta$.

Proposition (Asymptotic confidence interval). If $\hat{\theta}_n$ is asymptotically normal and $\widehat{\text{se}}$ is a consistent estimator of $\text{se}(\hat{\theta}_n)$, then

$$C_n = \left(\hat{\theta}_n - z_{\alpha/2}\,\widehat{\text{se}},\ \ \hat{\theta}_n + z_{\alpha/2}\,\widehat{\text{se}}\right)$$

satisfies $P_\theta(\theta \in C_n) \to 1 - \alpha$, where $z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2)$.

Proof. Write $P_\theta(\theta \in C_n) = P\left(\left|(\hat{\theta}_n - \theta)/\widehat{\text{se}}\right| \leq z_{\alpha/2}\right)$. Since $(\hat{\theta}_n - \theta)/\text{se}(\hat{\theta}_n) \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$ and $\widehat{\text{se}}/\text{se}(\hat{\theta}_n) \xrightarrow{p} 1$, Slutsky's lemma gives $(\hat{\theta}_n - \theta)/\widehat{\text{se}} \xrightarrow{d} Z$. Hence the probability converges to $P(|Z| \leq z_{\alpha/2}) = 1 - \alpha$. □

Chapter 5

Nonparametric Inference

5.1 Estimating the CDF and Functionals of the CDF

5.1.1 The Empirical CDF

Definition (Empirical CDF). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$. The **empirical CDF** is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(X_i \leq x).$$

Theorem. At any fixed $x \in \mathbb{R}$:

$$E(\hat{F}_n(x)) = F(x), \qquad \mathrm{Var}(\hat{F}_n(x)) = \frac{F(x)(1 - F(x))}{n} \to 0,$$

and consequently $\hat{F}_n(x) \xrightarrow{p} F(x)$.

Proof. Each $\mathbf{1}(X_i \leq x) \sim \mathrm{Bern}(F(x))$, so $E(\hat{F}_n(x)) = F(x)$ and $\mathrm{Var}(\hat{F}_n(x)) = F(x)(1 - F(x))/n$. Since MSE $\to 0$, convergence in probability follows. □

Theorem (Glivenko–Cantelli). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$. Then

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

Theorem (Dvoretzky–Kiefer–Wolfowitz (DKW)). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$. For any $\varepsilon > 0$,

$$P\left(\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| > \varepsilon \right) \leq 2e^{-2n\varepsilon^2}.$$

Corollary (Confidence band for F). Set $\varepsilon_n = \sqrt{\frac{1}{2n} \log(2/\alpha)}$, and define

$$L(x) = \max(\hat{F}_n(x) - \varepsilon_n,\, 0), \quad U(x) = \min(\hat{F}_n(x) + \varepsilon_n,\, 1).$$

Then $P(L(x) \leq F(x) \leq U(x) \text{ for all } x) \geq 1 - \alpha$.

Proof. Direct inversion of the DKW bound: $P(\sup_x |\hat{F}_n(x) - F(x)| \leq \varepsilon_n) \geq 1 - 2e^{-2n\varepsilon_n^2} = 1 - \alpha$. □

5.1.2 Plug-in Estimators

Definition (Functional and plug-in estimator). A **functional** of F is any function $T(F)$. The **plug-in estimator** of $\theta = T(F)$ is $\hat{\theta}_n = T(\hat{F}_n)$.

A **linear functional** has the form $T(F) = \int r(x)\,\mathrm{d}F(x)$, and its plug-in estimator is

$$T(\hat{F}_n) = \int r(x)\,\mathrm{d}\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} r(X_i).$$

5.1.3 Functional Delta Method

To handle nonlinear functionals we extend the derivative to functions via the Gâteaux derivative.

Definition (Gâteaux derivative and influence function). The **Gâteaux derivative** of T at F in direction G is

$$L_F(G) = \lim_{\varepsilon \to 0} \frac{T((1-\varepsilon)F + \varepsilon G) - T(F)}{\varepsilon}.$$

When $G = \delta_x$ (point mass at x), the **influence function** is $L_F(x) := L_F(\delta_x)$.

Proposition (Properties of linear functionals). If $T(F) = \int a(x)\,\mathrm{d}F(x)$, then:

1. $L_F(x) = a(x) - T(F)$.
2. For any G: $L_F(G) = T(G) - T(F)$, equivalently $T(G) = T(F) + \int L_F(x)\,\mathrm{d}G(x)$.
3. $\int L_F(x)\,\mathrm{d}F(x) = 0$.

Theorem (Functional CLT for linear functionals). Let T be a linear functional with $\tau^2 = \int L_F^2(x)\,\mathrm{d}F(x) < \infty$. Then

$$\sqrt{n}(T(\hat{F}_n) - T(F)) \xrightarrow{d} \mathcal{N}(0, \tau^2).$$

The variance τ^2 is consistently estimated by $\hat{\tau}^2 = \frac{1}{n}\sum_{i=1}^{n} \hat{L}^2(X_i)$, where $\hat{L}(x) = a(x) - T(\hat{F}_n)$.

Proof. By property (2) of Proposition 5.1.3 with $G = \hat{F}_n$:

$$T(\hat{F}_n) - T(F) = \int L_F(x)\,\mathrm{d}\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} L_F(X_i).$$

By property (3), $E[L_F(X_i)] = 0$, so the sum has mean zero and variance τ^2/n. The CLT gives asymptotic normality. Consistency of $\hat{\tau}^2$ follows from the law of large numbers. $\square$

5.2 Bootstrap Methods

5.2.1 Bootstrap Variance Estimation

Let $T_n = g(X_1, \ldots, X_n)$ be a statistic with unknown distribution F. The goal is to estimate $\mathrm{Var}_F(T_n)$.

Definition (Nonparametric bootstrap).

1. Estimate F by the empirical CDF $\hat{F}_n$.
2. Draw bootstrap samples $X_1^*, \ldots, X_n^* \overset{\text{iid}}{\sim} \hat{F}_n$ (equivalently, sample with replacement from $X_1, \ldots, X_n$).
3. For each bootstrap replicate $b = 1, \ldots, B$, compute $T_{n,b}^* = g(X_{1,b}^*, \ldots, X_{n,b}^*)$.

The **bootstrap variance estimate** is

$$v_{\text{boot}} = \frac{1}{B} \sum_{b=1}^{B} \left(T_{n,b}^* - \overline{T}_n^* \right)^2, \qquad \overline{T}_n^* = \frac{1}{B} \sum_{b=1}^{B} T_{n,b}^*.$$

5.2.2 Bootstrap Confidence Intervals

Let $\theta = T(F)$ and $\hat{\theta}_n = T(\hat{F}_n)$.

Definition (Normal interval).

$$\hat{\theta}_n \pm z_{\alpha/2} \sqrt{v_{\text{boot}}}.$$

Appropriate only when T_n is approximately normal.

Definition (Pivotal (basic) interval). Define the pivot $R_n = \hat{\theta}_n - \theta$. Its bootstrap estimate is $R_{n,b}^* = \hat{\theta}_{n,b}^* - \hat{\theta}_n$. Let r_q^* denote the q-th quantile of $(R_{n,1}^*, \ldots, R_{n,B}^*)$, and θ_q^* the q-th quantile of $(\hat{\theta}_{n,1}^*, \ldots, \hat{\theta}_{n,B}^*)$. The **pivotal confidence interval** is

$$\hat{C}_n = \left(2\hat{\theta}_n - \theta_{1-\alpha/2}^*, \;\; 2\hat{\theta}_n - \theta_{\alpha/2}^* \right).$$

Definition (Percentile interval).

$$C_n = \left(\theta_{\alpha/2}^*, \; \theta_{1-\alpha/2}^* \right).$$

Theorem. Under weak regularity conditions on $T(F)$, the pivotal interval satisfies $P_F(T(F) \in \hat{C}_n) \to 1 - \alpha$ as $n \to \infty$.

Proof sketch. If $H(r) = P(R_n \leq r)$ were known, the exact interval would use $H^{-1}(\alpha/2)$ and $H^{-1}(1 - \alpha/2)$. Bootstrap consistency (Theorem 5.2.2 relies on Class 8 results) shows $\hat{H}(r) \xrightarrow{p} H(r)$ uniformly, so estimated quantiles converge to the true ones. □

5.3 Consistency of Bootstrap Confidence Intervals

For n i.i.d. observations $X_1, \ldots, X_n$ in $\mathbb{R}^d$, the **empirical measure** is $\hat{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i}(A)$. Sampling from $\hat{P}_n$ is sampling with replacement from $X_1, \ldots, X_n$.

Lemma. Let F_n be a sequence of CDFs and F a CDF. Then $F_n^{-1}(p) \to F^{-1}(p)$ at every continuity point p of F^{-1} if and only if $F_n(x) \to F(x)$ at every continuity point x of F. Equivalently, $F_n \xrightarrow{d} F$.

Let $\hat{F}_n(x) = P\Big((\hat{\theta}_n^* - \hat{\theta}_n)/\sigma_n^* \le x \mid \hat{P}_n\Big)$, and define $\hat{\xi}_{n,1-\alpha} = \hat{F}_n^{-1}(1-\alpha)$. The bootstrap confidence interval is

$$\hat{I}_n(\alpha, \beta) = \left[\hat{\theta}_n - \hat{\xi}_{n,\beta}\hat{\sigma}_n,\ \ \hat{\theta}_n - \hat{\xi}_{n,1-\alpha}\hat{\sigma}_n\right].$$

Theorem (Consistency of bootstrap intervals). Suppose $(\hat{\theta}_n - \theta)/\hat{\sigma}_n \xrightarrow{d} T$, and that for almost every sequence $X_1, X_2, \ldots$, $(\hat{\theta}_n^* - \hat{\theta}_n)/\sigma_n^* \xrightarrow{d} T$ conditionally on $X_1, X_2, \ldots$, where T has a continuous CDF F. Then

$$P(\theta \in \hat{I}_n(\alpha, \beta)) \to 1 - \alpha - \beta.$$

Proof. By assumption, $\hat{F}_n \xrightarrow{d} F$ almost surely. By Lemma 5.3, $\hat{\xi}_{n,1-\alpha} = \hat{F}_n^{-1}(1-\alpha) \to F^{-1}(1-\alpha)$ a.s. (at continuity points of F^{-1}). By Slutsky's lemma,

$$(\hat{\theta}_n - \theta)/\hat{\sigma}_n - \hat{\xi}_{n,1-\alpha} \xrightarrow{d} T - F^{-1}(1-\alpha).$$

Therefore,

$$P(\theta \ge \hat{\theta}_n - \hat{\sigma}_n\hat{\xi}_{n,1-\alpha}) = P\left(\frac{\hat{\theta}_n - \theta}{\hat{\sigma}_n} \le \hat{\xi}_{n,1-\alpha}\right) \to P(T \le F^{-1}(1-\alpha)) = 1 - \alpha.$$

The same argument at level β gives $P(\theta \le \hat{\theta}_n - \hat{\sigma}_n\hat{\xi}_{n,\beta}) \to 1 - \beta$. Combining gives the result. □

5.4 Bootstrap Validity for the Sample Mean

Theorem. Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then conditionally on $X_1, X_2, \ldots$, for almost every sequence,

$$\sqrt{n}(\overline{X}_n^* - \overline{X}_n) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Proof. Let $Y_{n,i} = X_{n,i}^*/\sqrt{n}$, so $\sqrt{n}\overline{X}_n^* = \sum_i Y_{n,i}$. We verify the **Lindeberg–Feller CLT** conditions conditionally on $\hat{P}_n$.

Conditional mean and variance. $E[X_i^* \mid \hat{P}_n] = \overline{X}_n \to \mu$ a.s., and

$$\sum_{i=1}^{n} \mathrm{Var}(Y_{n,i} \mid \hat{P}_n) = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}_n^2 \xrightarrow{a.s.} \sigma^2.$$

Lindeberg condition. For fixed $\varepsilon > 0$ and $M \le \sqrt{n}\varepsilon$:

$$\sum_{i=1}^{n} E\Big[\|Y_{n,i}\|^2 \mathbf{1}\{\|Y_{n,i}\| > \varepsilon\} \mid \hat{P}_n\Big] \le \frac{1}{n}\sum_{i=1}^{n} \|X_i\|^2 \mathbf{1}\{\|X_i\| > M\} \xrightarrow{a.s.} E[\|X\|^2 \mathbf{1}\{\|X\| > M\}] \to 0$$

as $M \to \infty$, since $E[X^2] < \infty$ implies uniform integrability.

By the Lindeberg–Feller CLT, $\sum_i (Y_{n,i} - E[Y_{n,i} \mid \hat{P}_n]) = \sqrt{n}(\overline{X}_n^* - \overline{X}_n) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ a.s. $\square$

5.5 Lindeberg–Feller CLT

Theorem (Lindeberg–Feller CLT). For each n, let $(Y_{n,1}, \ldots, Y_{n,k_n})$ be independent with finite variances. Suppose for every $\varepsilon > 0$,

$$\sum_{i=1}^{k_n} E[\|Y_{n,i}\|^2 \mathbf{1}\{\|Y_{n,i}\| > \varepsilon\}] \to 0 \quad \text{as } n \to \infty,$$

and $\sum_{i=1}^{k_n} \mathrm{Var}(Y_{n,i}) \to \sigma^2$. Then

$$\sum_{i=1}^{k_n} (Y_{n,i} - E[Y_{n,i}]) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Chapter 6

Parametric Inference

6.1 Parametric Frequentist Inference

6.1.1 Maximum Likelihood Estimation

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x;\theta)$ with $\theta \in \Theta \subseteq \mathbb{R}^k$.

Definition (Likelihood and MLE). The **likelihood** is $L_n(\theta) = \prod_{i=1}^{n} f(X_i;\theta)$. The **log-likelihood** is $\ell_n(\theta) = \log L_n(\theta)$. The **maximum likelihood estimator (MLE)** is

$$\hat{\theta}_n = \arg\max_{\theta\in\Theta} L_n(\theta) = \arg\max_{\theta\in\Theta} \ell_n(\theta).$$

6.1.2 Consistency of the MLE

Definition (KL divergence and identifiability). The **Kullback–Leibler (KL) divergence** between $f(\cdot;\theta)$ and $f(\cdot;\theta')$ is

$$D(\theta,\theta') = \int f(x;\theta) \log \frac{f(x;\theta)}{f(x;\theta')}\,\mathrm{d}x \geq 0,$$

with $D(\theta,\theta') = 0$ iff $f(\cdot;\theta) = f(\cdot;\theta')$ a.e. The model is **identifiable** if $\theta \neq \theta'$ implies $D(\theta,\theta') > 0$.

Maximizing $\ell_n(\theta)$ is equivalent to maximizing

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log \frac{f(X_i;\theta)}{f(X_i;\theta^*)}.$$

By the LLN, $M_n(\theta) \xrightarrow{p} -D(\theta^*,\theta) =: M(\theta)$ pointwise, and $M(\theta^*) = 0 \geq M(\theta)$ for all θ by non-negativity of KL.

Theorem (Consistency of the MLE). Suppose the model is identifiable and $\sup_\theta |M_n(\theta) - M(\theta)| \xrightarrow{p} 0$. Then $\hat{\theta}_n \xrightarrow{p} \theta^*$.

Proof. The uniform convergence assumption gives $M(\hat{\theta}_n) - M(\theta^*) \leq |M_n(\hat{\theta}_n) - M(\hat{\theta}_n)| + |M_n(\theta^*) - M(\theta^*)| \xrightarrow{p} 0$. So $M(\hat{\theta}_n) \xrightarrow{p} M(\theta^*)$. By identifiability, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $\|\theta - \theta^*\| \geq \varepsilon \Rightarrow M(\theta) < M(\theta^*) - \delta$. Therefore

$$P(\|\hat{\theta}_n - \theta^*\| \geq \varepsilon) \leq P(M(\hat{\theta}_n) < M(\theta^*) - \delta) \to 0.$$

□

6.1.3 Equivariance

Theorem (Equivariance of the MLE). Let $\tau = g(\theta)$ for any function g. Define the induced likelihood $L_n^*(\tau) = \sup_{\theta: g(\theta)=\tau} L_n(\theta)$. Then the MLE of τ is $\hat{\tau}_n = g(\hat{\theta}_n)$.

6.1.4 Asymptotic Normality

Definition (Score and Fisher information). The **score function** is $s(X;\theta) = \nabla_\theta \ell(\theta)$. The **Fisher information matrix** is

$$[I_n(\theta)]_{i,j} = E\left[\frac{\partial \ell_n(\theta)}{\partial \theta_i}\frac{\partial \ell_n(\theta)}{\partial \theta_j}\right] = -E\left[\frac{\partial^2 \ell_n(\theta)}{\partial \theta_i \partial \theta_j}\right].$$

Proposition. $E_\theta[s(X;\theta)] = 0$, and $[I(\theta)]_{i,j} = \mathrm{Cov}(s(X;\theta)_i,\, s(X;\theta)_j)$.

Proof. Differentiate $\int f(x;\theta)\,\mathrm{d}x = 1$ with respect to θ_i (under the integral):

$$0 = \int \frac{\partial}{\partial \theta_i} f(x;\theta)\,\mathrm{d}x = \int \frac{\partial \log f(x;\theta)}{\partial \theta_i} f(x;\theta)\,\mathrm{d}x = E[s(X;\theta)_i].$$

Then $[I(\theta)]_{i,j} = E[s_i s_j] - E[s_i]E[s_j] = \mathrm{Cov}(s_i, s_j)$. □

Theorem (Asymptotic normality of the MLE). Under regularity conditions,

$$I_n(\hat{\theta}_n)^{1/2}(\hat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, I_k),$$

and $\mathrm{Cov}(\hat{\theta}_n^i, \hat{\theta}_n^j) \approx [I_n(\hat{\theta}_n)^{-1}]_{i,j}$.

Proof for $\Theta \subseteq \mathbb{R}$. Taylor-expand $\ell_n'(\theta^*)$ around $\hat{\theta}_n$ (where $\ell_n'(\hat{\theta}_n) = 0$):

$$0 = \ell_n'(\hat{\theta}_n) = \ell_n'(\theta^*) + (\hat{\theta}_n - \theta^*)\ell_n''(\theta^*) + \cdots$$

Rearranging:

$$\sqrt{n}(\hat{\theta}_n - \theta^*) = -\frac{\ell_n'(\theta^*)/\sqrt{n}}{\ell_n''(\theta^*)/n}.$$

Numerator: $-\ell_n'(\theta^*)/\sqrt{n} = \frac{1}{\sqrt{n}}\sum_{i=1}^n(-s(X_i;\theta^*))$. Since $E[s] = 0$ and $\mathrm{Var}(s) = I(\theta^*)$, the CLT gives $-\ell_n'(\theta^*)/\sqrt{n} \xrightarrow{d} \mathcal{N}(0, I(\theta^*))$.

Denominator: $\ell_n''(\theta^*)/n = \dfrac{1}{n}\sum_i \partial^2 \log f(X_i;\theta^*)/\partial\theta^2 \xrightarrow{p} -I(\theta^*)$ by the LLN.

Combining by Slutsky's lemma:

$$\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, 1/I(\theta^*)).$$

Since $I(\hat{\theta}_n) \xrightarrow{p} I(\theta^*)$ by continuity and consistency, we may replace θ^* with $\hat{\theta}_n$. □

6.2 Parametric Bayesian Inference

6.2.1 The Bayesian Method

Definition (Bayesian inference). Given a prior $f(\theta)$ and likelihood $f(x \mid \theta)$, the **posterior distribution** after observing $x^n = (x_1, \ldots, x_n)$ is

$$f(\theta \mid x^n) = \frac{f(\theta)\, f(x^n \mid \theta)}{\int f(\theta)\, f(x^n \mid \theta)\, \mathrm{d}\theta} \propto L_n(\theta)\, f(\theta).$$

Definition (Posterior predictive). The **posterior predictive distribution** for a new observation X_{n+1} is

$$f(x_{n+1} \mid x^n) = \int f(x_{n+1} \mid \theta)\, f(\theta \mid x^n)\, \mathrm{d}\theta.$$

Definition (MAP estimator). The **maximum a posteriori (MAP)** estimator is $\hat{\theta}_{\mathrm{MAP}} = \arg\max_\theta f(\theta \mid x^n)$.

Theorem (Bernstein–von Mises). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_{\theta^*}$. Let $\hat{\theta}_n$ be the MLE and $\widehat{\mathrm{se}} = 1/\sqrt{n\, I(\hat{\theta}_n)}$. Under regularity conditions, the posterior is approximately $\mathcal{N}(\hat{\theta}_n, \widehat{\mathrm{se}}^2)$, and if C_n is an approximate $1-\alpha$ confidence interval,

$$P(\theta \in C_n \mid X^n) \to 1 - \alpha.$$

The Bernstein–von Mises theorem says that for large n, the likelihood overwhelms the prior (as long as the prior assigns positive mass near θ^*), and Bayesian and frequentist procedures agree to first order.

6.2.2 Prior Specification

Definition (Flat (uniform) prior). A **flat prior** on Θ sets $f(\theta) \propto 1$. If Θ is not compact this may be **improper** (not integrable), yet can still yield a proper posterior.

Remark. Flat priors are not parameterization-invariant. If $\psi = h(\theta)$ is a reparameterization, a flat prior on θ becomes $f_\Psi(\psi) \propto |h^{-1\prime}(\psi)|$, which is generally not flat.

Definition (Jeffreys prior). The **Jeffreys prior** is

$$f(\theta) \propto \det(I(\theta))^{1/2},$$

where $I(\theta)$ is the Fisher information matrix. It is invariant under reparameterization.

Proposition (Reparameterization invariance of Jeffreys prior). Let $\psi = h(\theta)$ for a one-to-one h with Jacobian J_h. Then

$$\det(I_\Psi(\psi))^{1/2} = \det(I(h^{-1}(\psi)))^{1/2} \cdot |\det(J_{h^{-1}}(\psi))|\,,$$

which is the transformation of $f(\theta) \propto \det(I(\theta))^{1/2}$ from θ to ψ. Hence specifying the Jeffreys prior directly in Ψ agrees with transforming from Θ.

Proof. By the chain rule for Fisher information under reparameterization, $I_\Psi(\psi) = J_{h^{-1}}(\psi)^\top I(h^{-1}(\psi))\, J_{h^{-1}}($ Taking determinants: $\det(I_\Psi(\psi)) = \det(I(h^{-1}(\psi))) \cdot \det(J_{h^{-1}}(\psi))^2$. Taking square roots and applying the standard change-of-variables formula for densities gives the result. □

6.3 Statistical Decision Theory

6.3.1 Risk and Loss

Definition (Loss function and risk). A **loss function** $L : \Theta \times \Theta \to \mathbb{R}$ measures the cost of estimating θ by $\hat{\theta}$. The **risk** of an estimator $\hat{\theta}$ is

$$R(\theta, \hat{\theta}) = E_\theta[L(\theta, \hat{\theta}(X))].$$

Under **squared error loss** $L(\theta, \hat{\theta}) = (\hat{\theta} - \theta)^2$, the risk equals the MSE:

$$R(\theta, \hat{\theta}) = \mathrm{Var}_\theta(\hat{\theta}) + \mathrm{bias}_\theta(\hat{\theta})^2.$$

6.3.2 Comparing Risk Functions

Since $R(\theta, \hat{\theta})$ is a function of θ, comparing two estimators requires reducing risk to a scalar.

Definition (Maximum risk and Bayes risk). The **maximum risk** is $\overline{R}(\hat{\theta}) = \sup_\theta R(\theta, \hat{\theta})$. For a prior $f(\theta)$, the **Bayes risk** is

$$r(f, \hat{\theta}) = \int R(\theta, \hat{\theta})\, f(\theta)\, \mathrm{d}\theta.$$

Definition (Minimax estimator). $\hat{\theta}$ is **minimax** if $\sup_\theta R(\theta, \hat{\theta}) = \inf_{\tilde{\theta}} \sup_\theta R(\theta, \tilde{\theta})$.

Definition (Bayes estimator). The **Bayes estimator** with respect to prior f minimizes the Bayes risk: $r(f, \hat{\theta}) = \inf_{\tilde{\theta}} r(f, \tilde{\theta})$.

6.3.3 Finding Bayes Estimators

Definition (Posterior risk). The **posterior risk** is $r(\hat{\theta} \mid x) = \int L(\theta, \hat{\theta}(x))\, f(\theta \mid x)\, \mathrm{d}\theta$.

Theorem. The Bayes risk satisfies $r(f, \hat{\theta}) = \int r(\hat{\theta} \mid x)\, m(x)\, \mathrm{d}x$, where $m(x) = \int f(x \mid \theta) f(\theta)\, \mathrm{d}\theta$. Consequently:

- Under **squared error loss**, the Bayes estimator is the **posterior mean**: $\hat{\theta}(x) = E[\theta \mid X = x]$.
- Under **absolute error loss** $L(\theta, \hat{\theta}) = |\theta - \hat{\theta}|$, the Bayes estimator is the **posterior median**.

Proof. $r(f, \hat{\theta}) = \int\int L(\theta, \hat{\theta}(x))\, f(\theta \mid x)\, \mathrm{d}\theta\, m(x)\, \mathrm{d}x = \int r(\hat{\theta} \mid x)\, m(x)\, \mathrm{d}x$. Since $m(x) \geq 0$, the Bayes estimator minimizes $r(\hat{\theta} \mid x)$ pointwise for m-almost every x. For squared error, $r(\hat{\theta} \mid x) = E[(\theta - \hat{\theta}(x))^2 \mid x]$, which is minimized at $\hat{\theta}(x) = E[\theta \mid x]$. □

6.3.4 Minimax Decision Rules

Theorem (Bayes ⇒ minimax via constant risk). Let $\hat{\theta}^f$ be a Bayes estimator under prior f. If $R(\theta, \hat{\theta}^f) \leq r(f, \hat{\theta}^f)$ for all $\theta \in \Theta$, then $\hat{\theta}^f$ is minimax and f is a **least favorable prior**.

Proof. Suppose $\hat{\theta}^f$ is not minimax. Then there exists $\hat{\theta}_0$ with $\sup_\theta R(\theta, \hat{\theta}_0) < \sup_\theta R(\theta, \hat{\theta}^f)$. Since the average is at most the supremum:

$$r(f, \hat{\theta}_0) \leq \sup_\theta R(\theta, \hat{\theta}_0) < \sup_\theta R(\theta, \hat{\theta}^f) \leq r(f, \hat{\theta}^f).$$

This contradicts $\hat{\theta}^f$ being Bayes under f. □

Corollary. If $\hat{\theta}$ is Bayes under some prior f and has *constant risk* $R(\theta, \hat{\theta}) = c$ for all θ, then $\hat{\theta}$ is minimax.

Proof. Constant risk gives $r(f, \hat{\theta}) = c = R(\theta, \hat{\theta})$ for all θ, so the condition of Theorem 6.3.4 holds. □

Chapter 7

Prediction and Model Selection

7.1 Empirical Risk Minimization and Cross-Validation

7.1.1 Prediction and Risk

Definition (Prediction risk). For pairs $(X, Y) \sim P$ and a prediction function f, the **risk** under loss L is

$$R(P, f) = E_P[L(Y, f(X))] = \int L(y, f(x))\, \mathrm{d}P(x, y).$$

Proposition (Optimal predictors). Under squared error loss with $(X_i, Y_i) \overset{\text{iid}}{\sim} P$ and $Y_i \in \mathbb{R}$, the minimizer of risk is $f^*(x) = E_P[Y \mid X = x]$.
Under 0-1 loss with $Y \in \{0, 1\}$, the minimizer is the **Bayes classifier**:

$$f^*(x) = \begin{cases} 1 & \text{if } P(Y = 1 \mid X = x) \geq \frac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

Proof for squared error. $R(P, f) = E[(Y - f(X))^2] = E[(Y - f^*(X))^2] + E[(f^*(X) - f(X))^2] \geq E[(Y - f^*(X))^2]$, using $E[(Y - f^*(X))(f^*(X) - f(X))] = 0$ by the tower property. □

7.1.2 Empirical Risk Minimization

Definition (Empirical risk and ERM). Given training data $\mathcal{T}_n = ((X_1, Y_1), \ldots, (X_n, Y_n)) \overset{\text{iid}}{\sim} P$, the **empirical risk** is

$$\hat{R}_n(f) = R(\hat{P}_n, f) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, f(X_i)).$$

Empirical risk minimization (ERM): $\hat{f}_n = \arg\min_{f \in \mathcal{F}} \hat{R}_n(f)$.

Definition (Generalization risk). For fixed training data $\mathcal{T}_n$, the **generalization risk** is

$$R(P, \hat{f} \mid \mathcal{T}_n) = E_P[L(Y^*, \hat{f}_n(X^*)) \mid \mathcal{T}_n],$$

where $(X^*, Y^*) \sim P$ is independent of $\mathcal{T}_n$. The (unconditional) risk is $R(P, \hat{f}) = E[R(P, \hat{f} \mid \mathcal{T}_n)]$.

Proposition (Empirical risk underestimates risk). Under squared error loss,

$$E[\hat{R}_n(\hat{f}_n)] - R(P, \hat{f}) = -\frac{2}{n} \sum_{i=1}^{n} E\Big[\mathrm{Cov}(\hat{f}_n(X_i), Y_i \mid X_i)\Big].$$

In particular, $E[\hat{R}_n(\hat{f}_n)] \leq R(P, \hat{f})$ (empirical risk is a downward-biased estimator of risk).

Proof. Condition on X_i and compare in-sample prediction $\hat{f}_n(X_i)$ to out-of-sample prediction at the same point. For a new $Y_i^* \sim P(Y \mid X_i)$ independent of $\mathcal{T}_n$:

$$E[(\hat{f}_n(X_i) - Y_i)^2 \mid X_i] - E[(\hat{f}_n(X_i) - Y_i^*)^2 \mid X_i] = -2\,\mathrm{Cov}(\hat{f}_n(X_i), Y_i \mid X_i),$$

since Y_i^* is conditionally independent of $\hat{f}_n(X_i)$ given X_i, while Y_i is not. Summing over i and taking expectations gives the result. □

7.1.3 Cross-Validation

Definition (Leave-one-out cross-validation (LOOCV)). Estimate $\hat{f}^{(i)}$ on all data except (X_i, Y_i). The **LOOCV error** is

$$\hat{R}_{\mathrm{LOOCV}}(\hat{f}) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, \hat{f}^{(i)}(X_i)).$$

Theorem (LOOCV for linear smoothers). For linear regression with design matrix X and hat matrix $U = X(X^\top X)^{-1}X^\top$, the LOOCV error under squared error loss equals

$$\hat{R}_{\mathrm{LOOCV}} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{Y_i - \hat{Y}_i}{1 - U_{ii}} \right)^2,$$

and can be computed from a single model fit.

Proof. The Sherman–Morrison–Woodbury identity shows that leaving out observation i changes $\hat{f}(X_i)$ to $\hat{f}^{(i)}(X_i) = \hat{Y}_i - U_{ii}(Y_i - \hat{Y}_i)/(1 - U_{ii})$. Therefore $Y_i - \hat{f}^{(i)}(X_i) = (Y_i - \hat{Y}_i)/(1 - U_{ii})$. □

Definition (k-fold cross-validation). Randomly partition observations into k roughly equal **folds**. For each fold j, train on the remaining $k - 1$ folds and evaluate loss on fold j. Average the k held-out errors. LOOCV is the special case $k = n$.

7.2 Nonparametric Curve Estimation

7.2.1 Setup and Risk

Let f be an unknown function (a PDF or a regression function) to be estimated from data. The estimator $\hat{f}_n$ is itself a random function. Performance is measured by the **integrated squared error (ISE)**:

$$L(f, \hat{f}) = \int (f(u) - \hat{f}(u))^2 \, \mathrm{d}u,$$

with corresponding **risk** $R(f, \hat{f}) = E[L(f, \hat{f})]$. Decomposing pointwise:

$$R(f, \hat{f}) = \int b^2(u) \, \mathrm{d}u + \int v(u) \, \mathrm{d}u,$$

where $b(u) = E[\hat{f}(u)] - f(u)$ is the bias and $v(u) = \mathrm{Var}(\hat{f}(u))$ is the variance.

7.2.2 Kernel Density Estimation

Definition (Kernel). A **kernel** is a smooth function K satisfying $K(x) \geq 0$, $\int K(x) \, \mathrm{d}x = 1$, $\int xK(x) \, \mathrm{d}x = 0$, and $\sigma_K^2 = \int x^2 K(x) \, \mathrm{d}x < \infty$.

Definition (Kernel density estimator (KDE)). With bandwidth $h > 0$,

$$\hat{f}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K\left(\frac{x - X_i}{h}\right).$$

Theorem (Risk of the KDE). Under regularity conditions on f, the risk of the KDE using bandwidth h_n is

$$R(f, \hat{f}_n) = \frac{1}{4} \sigma_K^4 h_n^4 \int (f''(x))^2 \, \mathrm{d}x + \frac{1}{nh_n} \int K^2(x) \, \mathrm{d}x + O(n^{-1}) + O(h_n^6).$$

Proof of bias term. Write $K_h(x, X) = K((x - X)/h)/h$. Then $E[\hat{f}_n(x)] = E[K_h(x, X)]$. Change variables $u = (t - x)/h$:

$$E[K_h(x, X)] = \int K(u) \, f(x - hu) \, \mathrm{d}u.$$

Taylor-expand $f(x - hu) = f(x) - huf'(x) + \frac{h^2 u^2}{2} f''(x) + \cdots$ and integrate. Using $\int K(u) \, \mathrm{d}u = 1$ and $\int uK(u) \, \mathrm{d}u = 0$ (kernel symmetry):

$$E[\hat{f}_n(x)] = f(x) + \frac{h^2}{2} \sigma_K^2 f''(x) + O(h^4).$$

So $b_n(x) = \frac{h_n^2}{2} \sigma_K^2 f''(x) + O(h_n^4)$, and $\int b_n^2(u) \, \mathrm{d}u = \frac{1}{4} \sigma_K^4 h_n^4 \int (f''(x))^2 \, \mathrm{d}x + O(h_n^6)$. $\square$

Proof of variance term. $\mathrm{Var}[\hat{f}_n(x)] = \mathrm{Var}[K_h(x, X)]/n$. Since $\mathrm{Var}[K_h] = E[K_h^2] - (E[K_h])^2$, and $E[K_h^2] = \dfrac{1}{h^2}\displaystyle\int K((x-t)/h)^2 f(t)\,\mathrm{d}t = \dfrac{f(x)}{h}\displaystyle\int K^2(u)\,\mathrm{d}u + O(1)$, we get $v_n(x) = \dfrac{f(x)\int K^2\,\mathrm{d}u}{nh_n} + O(n^{-1})$. □

Corollary (Optimal bandwidth). The asymptotically optimal bandwidth minimizing the leading terms of risk is

$$h^* = \left(\frac{\int K^2(x)\,\mathrm{d}x}{n\,\sigma_K^4 \int (f''(x))^2\,\mathrm{d}x}\right)^{1/5},$$

giving optimal risk $R^*(f, \hat{f}_n) = O(n^{-4/5})$.

Proof. The leading terms of risk are $\dfrac{1}{4}\sigma_K^4 h^4 C_1 + \dfrac{C_2}{nh}$ where $C_1 = \displaystyle\int (f'')^2$ and $C_2 = \displaystyle\int K^2$. Differentiating with respect to h and setting to zero: $\sigma_K^4 h^3 C_1 = C_2/(nh^2)$, giving $h^5 = C_2/(n\sigma_K^4 C_1)$. Substituting back, risk $\propto n^{-4/5}$. □

Remark. In practice h is chosen by cross-validation using the leave-one-out criterion

$$\hat{J}(\hat{f}) = \int \hat{f}(x)^2\,\mathrm{d}x - \frac{2}{n}\sum_{i=1}^{n} \hat{f}_{-i}(X_i),$$

where $\hat{f}_{-i}$ is the KDE excluding X_i.

7.2.3 Nonparametric Regression

Definition (Regression function). $r(x) = E(Y \mid X = x)$.

Definition (Nadaraya–Watson kernel estimator (NWKE)).

$$\hat{r}(x) = \sum_{i=1}^{n} w_i(x)\,Y_i, \qquad w_i(x) = \frac{K_h(x, x_i)}{\sum_{j=1}^{n} K_h(x, x_j)}.$$

Proposition. The NWKE equals the plug-in estimator $\hat{r}(x) = \displaystyle\int y\,\hat{f}(y \mid x)\,\mathrm{d}y$, where $\hat{f}(y \mid x) = \hat{f}(x, y)/\hat{f}(x)$ uses product-kernel KDEs.

Remark. Under the optimal bandwidth sequence $h^* = O(n^{-1/5})$, the risk of the NWKE also converges at rate $O(n^{-4/5})$.

7.3 Nonparametric Regression: Linear Smoothers and Basis Functions

7.3.1 Linear Smoothers

Definition (Linear smoother). An estimator $\hat{r}_n$ of r is a **linear smoother** if for each x there exists a weight vector $\ell(x) = (\ell_1(x), \ldots, \ell_n(x))^\top$ such that

$$\hat{r}_n(x) = \sum_{i=1}^{n} \ell_i(x)\, Y_i.$$

The vector of fitted values is $\hat{\mathbf{r}}_n = LY$, where L is the $n \times n$ **smoothing matrix** with $L_{ij} = \ell_j(x_i)$.
The **effective degrees of freedom** is $\nu_L = \mathrm{tr}(L)$.

The NWKE is a linear smoother with $\ell_i(x) = w_i(x)$ from (13.17).

Definition (Leave-one-out CV for linear smoothers). For linear smoothers satisfying $\sum_i \ell_i(x) = 1$, the leave-one-out CV score

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{r}^{(-i)}(x_i))^2$$

can be computed from a single fit as

$$\hat{R}(h) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{Y_i - \hat{r}_n(x_i)}{1 - \ell_i(x_i)} \right)^2.$$

Proof. When observation i is left out, the remaining weights $\ell_j(x)$ for $j \neq i$ are renormalized to sum to 1:

$$\hat{r}^{(-i)}(x) = \sum_{j \neq i} Y_j\, \ell_{j,(-i)}(x), \quad \ell_{j,(-i)}(x) = \frac{\ell_j(x)}{\sum_{k \neq i} \ell_k(x)} = \frac{\ell_j(x)}{1 - \ell_i(x)}.$$

Therefore

$$\hat{r}^{(-i)}(x_i) = \frac{\sum_{j \neq i} Y_j\, \ell_j(x_i)}{1 - \ell_i(x_i)} = \frac{\hat{r}_n(x_i) - Y_i\, \ell_i(x_i)}{1 - \ell_i(x_i)},$$

and $Y_i - \hat{r}^{(-i)}(x_i) = (Y_i - \hat{r}_n(x_i))/(1 - \ell_i(x_i))$. □

7.3.2 Regularization and Splines

Definition (Penalized least squares). Minimize the **penalized squared error**

$$M(\lambda) = \sum_{i=1}^{n} (Y_i - \hat{r}_n(x_i))^2 + \lambda J(r),$$

where the **roughness penalty** is $J(r) = \displaystyle\int (r''(x))^2\, \mathrm{d}x$.

As $\lambda \to \infty$, $r'' = 0$ forces linear regression. As $\lambda \to 0$, $\hat{r}$ interpolates the data.

Definition (Cubic spline). A **cubic spline** on knots $a \le \xi_1 < \cdots < \xi_k \le b$ is a function r that is a cubic polynomial on each interval (ξ_j, ξ_{j+1}) and has continuous first and second derivatives at each knot. A **natural spline** is additionally linear beyond the boundary points a, b.

Theorem (Smoothing spline). The minimizer of $M(\lambda)$ in (over all sufficiently smooth functions) is a natural cubic spline with knots at the data points $x_1, \ldots, x_n$.

Definition (B-spline basis and smoothing spline estimator). With $p = n+4$ B-spline basis functions $B_1(x), \ldots, B_p(x)$, model $r(x) = \sum_j \beta_j B_j(x)$. Define the $n \times p$ matrix $\mathbb{B}$ with $\mathbb{B}_{ij} = B_j(x_i)$ and the $p \times p$ penalty matrix $\Omega_{jk} = \int B_j''(x) B_k''(x)\,\mathrm{d}x$. The minimizer of $M(\lambda)$ is

$$\hat{\beta} = (\mathbb{B}^\top \mathbb{B} + \lambda\Omega)^{-1} \mathbb{B}^\top Y.$$

Proof. Substituting $r(x) = \mathbb{B}\beta$, the objective becomes $\|Y - \mathbb{B}\beta\|^2 + \lambda\beta^\top\Omega\beta$. Differentiating with respect to β and setting to zero: $-2\mathbb{B}^\top(Y - \mathbb{B}\beta) + 2\lambda\Omega\beta = 0$, giving $(\mathbb{B}^\top\mathbb{B} + \lambda\Omega)\beta = \mathbb{B}^\top Y$. □

Proposition. The smoothing spline estimator is a linear smoother with smoothing matrix

$$L = \mathbb{B}(\mathbb{B}^\top\mathbb{B} + \lambda\Omega)^{-1}\mathbb{B}^\top.$$

Proof. The fitted values are $\hat{\mathbf{r}}_n = \mathbb{B}\hat{\beta} = \mathbb{B}(\mathbb{B}^\top\mathbb{B} + \lambda\Omega)^{-1}\mathbb{B}^\top Y = LY$. □

7.4 Shrinkage and Regularization

7.5 The James–Stein Estimator, Shrinkage, and Ridge Regression

7.5.1 The Normal Means Model

Definition (Normal means model). Observe $X_1, \ldots, X_n$ independently with $X_i \sim \mathcal{N}(\theta_i, 1)$. The goal is to estimate $\theta^n = (\theta_1, \ldots, \theta_n)$.

The MLE is $\hat{\theta}_i = X_i$, with risk under squared error loss:

$$E_{\theta^n}\left[\|\hat{\theta}^n - \theta^n\|^2\right] = n.$$

Definition (Bayes estimator under a shared prior). Assume additionally $\theta_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. The marginal distribution of X_i is $X_i \sim \mathcal{N}(0, 1+\sigma^2)$. The posterior is

$$\theta_i \mid X_i \sim \mathcal{N}(\tau X_i,\ \tau), \quad \tau = \frac{\sigma^2}{1+\sigma^2},$$

so the Bayes estimator is $\hat{\theta}_i^b = \tau X_i$.

Proposition (Risk of the Bayes estimator). $E\left[\sum_{i=1}^{n}(\hat{\theta}_i^b - \theta_i)^2\right] = n\tau$, where the expectation is over both X^n and θ^n. Since $\tau < 1$, the Bayes estimator has strictly lower risk than the MLE.

7.5.2 The James–Stein Estimator

Since τ is unknown, we estimate it. Marginally $S_n = \sum_i X_i^2 \sim (1+\sigma^2)\chi_n^2$, so $\hat{\tau} = 1 - (n-2)/S_n$ is an unbiased estimate of τ (for $n > 2$). This gives the **James–Stein (JS) estimator**:

$$\hat{\theta}^{\mathrm{JS}} = \hat{\tau}\, X^n = \left(1 - \frac{n-2}{S_n}\right) X^n.$$

Theorem (James–Stein). Let $X_i \sim \mathcal{N}(\theta_i, 1)$ independently for $i = 1, \ldots, n$, and suppose $n > 2$. Then

$$E\left[\sum_{i=1}^{n}(\hat{\theta}_i^{\mathrm{JS}} - \theta_i)^2\right] \le 2 + \frac{(n-2)\sum_{i=1}^{n}\theta_i^2}{n-2+\sum_{i=1}^{n}\theta_i^2} < n,$$

so the JS estimator has strictly lower risk than the MLE for all θ^n.

Remark. The risk of the JS estimator is $E\left[\|\hat{\theta}^{\mathrm{JS}} - \theta^n\|^2\right] = n\tau + 2(1-\tau)$, which is strictly less than n for $n > 2$. Importantly, the prior on θ is *not* needed for this improvement; it is a property of the estimator under any fixed θ^n. The proof uses SURE (Class 16).

7.5.3 Ridge Regression

In linear regression with many covariates, one encodes a prior belief that most $\beta_j \approx 0$ via $\beta_j \sim \mathcal{N}(0, \lambda^{-1/2})$ and the model $Y_i = X_i\beta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. The posterior is proportional to

$$\exp\left(-\frac{1}{2\sigma^2}\|Y - X\beta\|^2 - \frac{\lambda}{2}\|\beta\|^2\right).$$

Definition (Ridge regression estimator).

$$\hat{\beta}(\lambda) = \arg\min_{\beta}\left\{\|Y - X\beta\|^2 + \frac{\lambda\sigma^2}{2}\|\beta\|^2\right\} = (X^\top X + \lambda\sigma^2 I)^{-1}X^\top Y.$$

Absorbing σ^2 into λ, this equals $(X^\top X + \lambda I)^{-1}X^\top X\,\hat{\beta}_{\mathrm{OLS}}$.

Proof. Differentiate $\|Y - X\beta\|^2 + \lambda\|\beta\|^2$ with respect to β and set to zero: $-2X^\top(Y - X\beta) + 2\lambda\beta = 0$, giving $(X^\top X + \lambda I)\beta = X^\top Y$. □

Remark. The tuning parameter λ controls shrinkage strength and is typically chosen by cross-validation.

7.6 Stein's Lemma, SURE, and James–Stein Analysis

7.6.1 Stein's Lemma

Lemma (Stein's lemma). Let $X \sim \mathcal{N}_k(\mu, \Sigma)$, and suppose $g : \mathbb{R}^k \to \mathbb{R}$ is (weakly) differentiable with $E[\|\nabla g(X)\|^2] < \infty$. Then

$$E[g(X)(X - \mu)] = \Sigma\, E[\nabla g(X)].$$

Proof for $k = 1$, $\mu = 0$, $\sigma^2 = 1$. Let φ be the standard normal PDF, so $\varphi'(z) = -z\varphi(z)$. Integration by parts:

$$E[g'(X)] = \int_{-\infty}^{\infty} g'(x)\varphi(x)\,\mathrm{d}x = \Big[g(x)\varphi(x)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} g(x)\varphi'(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} x\, g(x)\varphi(x)\,\mathrm{d}x = E[Xg(X)].$$

The boundary term vanishes because $E[|g'(X)|] < \infty$ implies $g(x)\varphi(x) \to 0$ as $x \to \pm\infty$. For $X \sim \mathcal{N}(\mu, \sigma^2)$, apply the result to $Z = (X - \mu)/\sigma$ and $f(z) = g(\sigma z + \mu)$. □

7.6.2 Stein's Unbiased Risk Estimate (SURE)

In the normal means model with $X_i \sim \mathcal{N}(\theta_i, 1)$, the risk of any estimator $\hat{\theta}^n = \hat{\theta}^n(X^n)$ under squared error loss is:

Theorem (SURE formula).

$$E\Big[\|\theta^n - \hat{\theta}^n\|^2\Big] = -n + E\Big[\|\hat{\theta}^n - X^n\|^2\Big] + 2\sum_{i=1}^{n} E\left[\frac{\partial \hat{\theta}_i}{\partial x_i}(X^n)\right].$$

Consequently, an unbiased estimate of risk is

$$\hat{R}(\hat{\theta}^n) = -n + \sum_{i=1}^{n} (\hat{\theta}_i - X_i)^2 + 2\sum_{i=1}^{n} \frac{\partial \hat{\theta}_i}{\partial x_i}(X^n).$$

Proof. Expand $\|\theta^n - \hat{\theta}^n\|^2 = \|\theta^n - X^n\|^2 + \|X^n - \hat{\theta}^n\|^2 + 2(\theta^n - X^n)^\top (X^n - \hat{\theta}^n)$ and take expectations. The first term gives $E[\|\theta^n - X^n\|^2] = n$. For the cross term, $E[(\theta_i - X_i)(\hat{\theta}_i - X_i)] = -\mathrm{Cov}(X_i, \hat{\theta}_i - X_i)$ (since θ_i is fixed). Applying Stein's lemma to $g_i(X^n) = \hat{\theta}_i(X^n) - X_i$ (and noting $\partial g_i/\partial x_i = \partial \hat{\theta}_i/\partial x_i - 1$):

$$E[X_i(\hat{\theta}_i - X_i)] = E\left[\frac{\partial \hat{\theta}_i}{\partial x_i} - 1\right], \quad \text{so} \quad E[(\theta_i - X_i)(\hat{\theta}_i - X_i)] = 1 - E\left[\frac{\partial \hat{\theta}_i}{\partial x_i}\right].$$

Summing over i and rearranging gives the formula. □

7.6.3 Linear Shrinkage and Oracle Risk

Consider estimators of the form $\hat{\theta}_i = bX_i$ for scalar b.

Proposition (Oracle risk of linear shrinkage). The risk-minimizing b (assuming θ^n is known) is $b^* = \|\theta^n\|^2/(n + \|\theta^n\|^2)$, giving oracle risk

$$\inf_{\hat{\theta}^n \in \mathcal{L}} R(\hat{\theta}^n, \theta^n) = \frac{n\|\theta^n\|^2}{n + \|\theta^n\|^2},$$

where $\mathcal{L}$ denotes the class of linear shrinkage estimators.

Proof. For $\hat{\theta}_i = bX_i$, $R = E[\|bX^n - \theta^n\|^2] = b^2 n + (1-b)^2\|\theta^n\|^2$. This is a convex quadratic in b; setting the derivative to zero gives $b^* = \|\theta^n\|^2/(n + \|\theta^n\|^2)$. □

Applying SURE to linear shrinkage ($\hat{\theta}_i = bX_i$, so $\partial\hat{\theta}_i/\partial x_i = b$) yields the data-driven estimate $b^* = 1 - n/S_n$, which is exactly $\hat{\tau}$ from the JS estimator.

Theorem (Risk bound for JS estimator).

$$R(\hat{\theta}_n^{\mathrm{JS}}, \theta^n) \le n - \frac{(n-2)^2}{n - 2 + \|\theta^n\|^2} = 2 + \frac{(n-2)\|\theta^n\|^2}{n - 2 + \|\theta^n\|^2}.$$

Remark. Combining with the oracle risk, this gives

$$\inf_{\hat{\theta}^n \in \mathcal{L}} R(\hat{\theta}^n, \theta^n) \le R(\hat{\theta}_n^{\mathrm{JS}}, \theta^n) \le 2 + \inf_{\hat{\theta}^n \in \mathcal{L}} R(\hat{\theta}^n, \theta^n),$$

so the JS estimator is within 2 of the best linear shrinkage estimator uniformly over all θ^n.

Chapter 8

General Asymptotic Theory

8.1 Consistency of M- and Z-estimators

8.1.1 M- and Z-estimators

Definition (M-estimator). Given i.i.d. $X_1, \ldots, X_n \sim P$ and a known function $m_\theta : \mathcal{X} \to \mathbb{R}$, define the **objective function**

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} m_\theta(X_i) = \hat{P}_n m_\theta.$$

An **M-estimator** $\hat{\theta}_n$ maximizes $M_n(\theta)$ over Θ. The population analogue is $M(\theta) = Pm_\theta = \int m_\theta(x)\,\mathrm{d}P(x)$. By the LLN, $M_n(\theta) \xrightarrow{p} M(\theta)$ pointwise.
Maximum likelihood is the special case $m_\theta = \log f_\theta$.

Definition (Z-estimator). When appropriate derivatives exist, maximizing M_n is equivalent to solving

$$\Psi_n(\theta) := \sum_{i=1}^{n} \psi_\theta(X_i) = 0,$$

for a vector-valued function ψ_θ. Any estimator obtained by solving such a system is a **Z-estimator**. For MLE, ψ_θ is the score function.
Throughout, θ_0 denotes the true parameter.

8.1.2 Consistency via Uniform Convergence

Theorem (Consistency of M-estimators). Let Θ be a metric space with metric d. Suppose:

1. **(ULLN)** $\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{p} 0$.

2. **(Well-separated maximum)** For every $\varepsilon > 0$, $\sup_{\theta:\, d(\theta,\theta_0) \geq \varepsilon} M(\theta) < M(\theta_0)$.

If $\hat{\theta}_n$ satisfies $M_n(\hat{\theta}_n) \geq M_n(\theta_0) - o_P(1)$, then $d(\hat{\theta}_n, \theta_0) \xrightarrow{p} 0$.

Proof. By assumption and the ULLN, $M_n(\theta_0) \xrightarrow{p} M(\theta_0)$, so $M_n(\hat{\theta}_n) \geq M(\theta_0) - o_P(1)$. Therefore

$$M(\theta_0) - M(\hat{\theta}_n) \leq M_n(\hat{\theta}_n) - M(\hat{\theta}_n) + o_P(1) \leq \sup_\theta |M_n(\theta) - M(\theta)| + o_P(1) \xrightarrow{p} 0.$$

By assumption (2), for every $\varepsilon > 0$ there exists $\delta > 0$ such that $d(\theta, \theta_0) \geq \varepsilon \Rightarrow M(\theta) < M(\theta_0) - \delta$. Hence

$$P(d(\hat{\theta}_n, \theta_0) \geq \varepsilon) \leq P(M(\hat{\theta}_n) < M(\theta_0) - \delta) \to 0. \qquad \square$$

Remark. Assumption (1) is a **uniform law of large numbers (ULLN)**. It holds when the function class $\{m_\theta : \theta \in \Theta\}$ is **Glivenko–Cantelli** (the empirical CDF example from Class 6 is the canonical instance). Verifying this in general requires tools from empirical process theory.
A Z-estimator $\hat{\theta}_n$ (zero of Ψ_n) also satisfies the theorem: a zero of Ψ_n maximizes $\theta \mapsto -\|\Psi_n(\theta)\|$, so set $m_\theta = -\|\psi_\theta\|$.

8.1.3 Consistency via Monotonicity

For scalar Θ, the ULLN can often be replaced by monotonicity.

Lemma (Consistency of Z-estimators via monotonicity). Let $\Theta \subseteq \mathbb{R}$. Suppose $\Psi_n(\theta) \xrightarrow{p} \Psi(\theta)$ for every $\theta \in \Theta$, and each $\theta \mapsto \Psi_n(\theta)$ is either:

1. continuous with exactly one zero $\hat{\theta}_n$; or
2. nondecreasing with $\Psi_n(\hat{\theta}_n) \xrightarrow{p} 0$.

Let θ_0 be a point with $\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon)$ for every $\varepsilon > 0$. Then $\hat{\theta}_n \xrightarrow{p} \theta_0$.

Proof. Case (1). If $\Psi_n(\theta_0 - \varepsilon) < 0$ and $\Psi_n(\theta_0 + \varepsilon) > 0$, then by continuity and the intermediate value theorem, $\hat{\theta}_n$ (the unique zero) satisfies $\theta_0 - \varepsilon < \hat{\theta}_n < \theta_0 + \varepsilon$. Since $\Psi_n(\theta_0 \pm \varepsilon) \xrightarrow{p} \Psi(\theta_0 \pm \varepsilon)$ and $\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon)$:

$$P(\theta_0 - \varepsilon < \hat{\theta}_n < \theta_0 + \varepsilon) \geq P(\Psi_n(\theta_0 - \varepsilon) < 0,\ \Psi_n(\theta_0 + \varepsilon) > 0) \to 1.$$

Case (2). If Ψ_n is nondecreasing and $\Psi_n(\hat{\theta}_n) \xrightarrow{p} 0$, then the same argument applies: $\Psi_n(\theta_0 - \varepsilon) < 0 < \Psi_n(\theta_0 + \varepsilon)$ with high probability implies $\hat{\theta}_n \in (\theta_0 - \varepsilon, \theta_0 + \varepsilon)$ with high probability. $\square$

Remark. The lemma also applies to nonincreasing $\tilde{\Psi}_n$ by considering $\Psi_n = -\tilde{\Psi}_n$.

Example (Consistency of the sample median). The sample median is a Z-estimator with $\psi_\theta(x) = \mathbf{1}(x > \theta) - \mathbf{1}(x \leq \theta) = \operatorname{sgn}(x - \theta)$. The map $\theta \mapsto \Psi_n(\theta)$ is nonincreasing, so apply Lemma 8.1.3 with $\Psi_n = -\sum_i \psi_\theta(X_i)$ (making it nondecreasing). The population function satisfies $\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon)$ whenever the population median θ_0 is unique, i.e., $P(X < \theta_0 - \varepsilon) < 1/2 < P(X > \theta_0 + \varepsilon)$ for all $\varepsilon > 0$. Hence the sample median $\xrightarrow{p} \theta_0$.

8.2 Asymptotic Normality of M- and Z-estimators

8.2.1 Asymptotic Normality of Z-estimators

Recall $\hat{\theta}_n$ solves $\Psi_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \psi_\theta(X_i) = \hat{P}_n\psi_\theta = 0$, and $P\psi_{\theta_0} = 0$.

Definition (Local Lipschitz condition). There exists a measurable $\overline{\psi}$ with $P(\overline{\psi}^2) < \infty$ such that for every θ_1, θ_2 in a neighborhood of θ_0 and each x,

$$\|\psi_{\theta_1}(x) - \psi_{\theta_2}(x)\| \leq \overline{\psi}(x)\|\theta_1 - \theta_2\|.$$

Theorem (Asymptotic normality of Z-estimators). For each θ in an open subset of $\mathbb{R}^k$, let $x \mapsto \psi_\theta(x)$ be measurable and satisfy the local Lipschitz condition. Assume $P\|\psi_{\theta_0}\|^2 < \infty$, and that $\theta \mapsto P\psi_\theta$ is differentiable at θ_0 with invertible derivative matrix V_{θ_0}. If $\hat{P}_n\psi_{\hat{\theta}_n} = o_P(n^{-1/2})$ and $\hat{\theta}_n \xrightarrow{p} \theta_0$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\theta_0}(X_i) + o_P(1),$$

which implies

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}_k\big(0,\ V_{\theta_0}^{-1}P(\psi_{\theta_0}\psi_{\theta_0}^\top)(V_{\theta_0}^{-1})^\top\big)\,.$$

Proof. **Step 1.** By the CLT, $\sqrt{n}(\hat{P}_n\psi_{\theta_0} - P\psi_{\theta_0}) \xrightarrow{d} \mathcal{N}_k(0, P(\psi_{\theta_0}\psi_{\theta_0}^\top))$.
Step 2. The local Lipschitz condition and consistency of $\hat{\theta}_n$ imply (via empirical process theory)

$$\sqrt{n}(\hat{P}_n\psi_{\hat{\theta}_n} - P\psi_{\hat{\theta}_n}) - \sqrt{n}(\hat{P}_n\psi_{\theta_0} - P\psi_{\theta_0}) \xrightarrow{p} 0. \tag{$*$}$$

Step 3. Since $\sqrt{n}\hat{P}_n\psi_{\hat{\theta}_n} = o_P(1)$ and $P\psi_{\theta_0} = 0$, the left side of $(*)$ equals

$$\sqrt{n}(P\psi_{\theta_0} - P\psi_{\hat{\theta}_n}) + o_P(1) = \sqrt{n}V_{\theta_0}(\theta_0 - \hat{\theta}_n) + o_P(1 + \sqrt{n}\|\theta_0 - \hat{\theta}_n\|),$$

using differentiability of $\theta \mapsto P\psi_\theta$ at θ_0.
Step 4. Combining with $(*)$:

$$\sqrt{n}(\hat{P}_n\psi_{\theta_0} - P\psi_{\theta_0}) + o_P(1) = \sqrt{n}V_{\theta_0}(\theta_0 - \hat{\theta}_n) + o_P(1 + \sqrt{n}\|\theta_0 - \hat{\theta}_n\|).$$

Step 5. We show $\sqrt{n}\|\theta_0 - \hat{\theta}_n\| = O_P(1)$ ($\hat{\theta}_n$ is $\sqrt{n}$-consistent). From Steps 3–4, rearranging gives $\sqrt{n}V_{\theta_0}(\hat{\theta}_n - \theta_0) = \sqrt{n}(\hat{P}_n\psi_{\theta_0} - P\psi_{\theta_0}) + o_P(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|)$. The right side converges in distribution (plus an error), so $\sqrt{n}\|\hat{\theta}_n - \theta_0\|$ is bounded in probability (since a convergent sequence is $O_P(1)$, and V_{θ_0} is invertible).
Step 6. With $\sqrt{n}\|\hat{\theta}_n - \theta_0\| = O_P(1)$, the error $o_P(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|) = o_P(O_P(1)) = o_P(1)$. Multiplying through by $V_{\theta_0}^{-1}$:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1}\sqrt{n}(\hat{P}_n\psi_{\theta_0} - P\psi_{\theta_0}) + o_P(1) = -V_{\theta_0}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\theta_0}(X_i) + o_P(1).$$

Asymptotic normality follows from Step 1 and Slutsky's lemma. □

8.2.2 Asymptotic Normality of M-estimators

Theorem (Asymptotic normality of M-estimators). For each θ in an open subset of $\mathbb{R}^k$, let $x \mapsto m_\theta(x)$ be measurable and satisfy the Lipschitz condition. Let $\theta \mapsto m_\theta(x)$ be differentiable at θ_0 for P-a.e. x with derivative $\dot{m}_\theta(x)$. Assume $\theta \mapsto Pm_\theta$ has a second-order Taylor expansion at its maximum θ_0:

$$Pm_\theta = Pm_{\theta_0} + \tfrac{1}{2}(\theta - \theta_0)^\top V_{\theta_0}(\theta - \theta_0) + o(\|\theta - \theta_0\|^2),$$

with V_{θ_0} invertible and symmetric negative definite. If $\hat{P}_n m_{\hat{\theta}_n} \geq \sup_\theta \hat{P}_n m_\theta - o_P(n^{-1})$ and $\hat{\theta}_n \xrightarrow{p} \theta_0$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{m}_{\theta_0}(X_i) + o_P(1) \xrightarrow{d} \mathcal{N}_k\big(0,\ V_{\theta_0}^{-1} P(\dot{m}_{\theta_0} \dot{m}_{\theta_0}^\top) V_{\theta_0}^{-1}\big)\,.$$

Proof sketch. Since $\sqrt{n}\|\hat{\theta}_n - \theta_0\| = O_P(1)$ (analogous to Z-case), set $\hat{h}_n = \sqrt{n}(\hat{\theta}_n - \theta_0) = O_P(1)$ and $\tilde{h}_n = -V_{\theta_0}^{-1}\sqrt{n}(\hat{P}_n \dot{m}_{\theta_0} - P\dot{m}_{\theta_0}) = O_P(1)$. Using the Taylor expansion and a technical lemma, for any $O_P(1)$ sequence h_n:

$$n\hat{P}_n(m_{\theta_0 + h_n/\sqrt{n}} - m_{\theta_0}) = \tfrac{1}{2} h_n^\top V_{\theta_0} h_n + \sqrt{n}(\hat{P}_n h_n^\top \dot{m}_{\theta_0} - P h_n^\top \dot{m}_{\theta_0}) + o_P(1).$$

Evaluating at $h_n = \hat{h}_n$ (LHS $\geq$ that at $h_n = \tilde{h}_n$ up to $o_P(1)$, since $\hat{\theta}_n$ approximately maximizes $\hat{P}_n m_\theta$) and completing the square shows

$$\tfrac{1}{2}(\hat{h}_n - \tilde{h}_n)^\top V_{\theta_0}(\hat{h}_n - \tilde{h}_n) + o_P(1) \geq 0.$$

Since V_{θ_0} is negative definite, this forces $\|\hat{h}_n - \tilde{h}_n\| \xrightarrow{p} 0$. Hence $\sqrt{n}(\hat{\theta}_n - \theta_0) = \tilde{h}_n + o_P(1) = -V_{\theta_0}^{-1}\sqrt{n}(\hat{P}_n \dot{m}_{\theta_0} - P\dot{m}_{\theta_0}) + o_P(1)$, and asymptotic normality follows from the CLT. □

Remark. For the MLE, $m_\theta = \log f_\theta$, $\dot{m}_{\theta_0} = s(X; \theta_0)$ (the score), and $V_{\theta_0} = -I(\theta_0)$, recovering the standard result $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$.

8.3 Step Estimators and Examples

8.3.1 Nonlinear Least Squares as a Z-estimator

Suppose $(X_1, Y_1), \ldots, (X_n, Y_n) \sim P_{\theta_0}$ follow $Y = f_{\theta_0}(X) + \varepsilon$ with $E[\varepsilon \mid X] = 0$, where f_θ is a parametric family. The least-squares objective is $\sum_i (Y_i - f_\theta(X_i))^2$, and assuming $\theta \mapsto f_\theta(x)$ is differentiable with $\dot{f}_\theta = \nabla_\theta f_\theta$, the corresponding Z-estimator solves

$$\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n (Y_i - f_\theta(X_i)) \dot{f}_\theta(X_i) = 0$$

Under suitable regularity (including the local Lipschitz condition on $\psi_\theta(x, y) = (y - f_\theta(x))\dot{f}_\theta(x)$) and assuming $E[\varepsilon^2 \mid X] = \sigma^2$ a.s., Theorem 20.1 gives

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\Big(0,\ \sigma^2 \big(E[\dot{f}_{\theta_0}(X) \dot{f}_{\theta_0}(X)^\top]\big)^{-1}\Big)\,.$$

8.3.2 One-Step Estimators

Definition (One-step estimator). Let $\tilde{\theta}_n$ be a $\sqrt{n}$-consistent preliminary estimator (not necessarily a Z-estimator). Suppose Ψ_n satisfies the following uniform differentiability condition: for every $M > 0$ and some nonsingular $\dot{\Psi}_0$,

$$\sup_{\sqrt{n}\|\theta-\theta_0\|<M} \left\| \sqrt{n}(\Psi_n(\theta) - \Psi_n(\theta_0)) - \dot{\Psi}_0\sqrt{n}(\theta - \theta_0) \right\| \xrightarrow{p} 0. \tag{†}$$

Let $\dot{\Psi}_{n,0} \xrightarrow{p} \dot{\Psi}_0$ be a sequence of nonsingular matrices. The **one-step estimator** is one Newton–Raphson step from $\tilde{\theta}_n$:

$$\hat{\theta}_n = \tilde{\theta}_n - \dot{\Psi}_{n,0}^{-1}\Psi_n(\tilde{\theta}_n).$$

Theorem (Asymptotic normality of one-step estimators). Let $\sqrt{n}\Psi_n(\theta_0) \xrightarrow{d} Z$. Suppose (†) holds and $\tilde{\theta}_n$ is $\sqrt{n}$-consistent. Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\dot{\Psi}_0^{-1}\sqrt{n}\Psi_n(\theta_0) + o_P(1) \xrightarrow{d} \mathcal{N}\Big(0,\ \dot{\Psi}_0^{-1}P(\psi_{\theta_0}\psi_{\theta_0}^\top)(\dot{\Psi}_0^{-1})^\top\Big).$$

Proof. Expand $\dot{\Psi}_{n,0}\sqrt{n}(\hat{\theta}_n - \theta_0)$:

$$\dot{\Psi}_{n,0}\sqrt{n}(\hat{\theta}_n - \theta_0) = \dot{\Psi}_{n,0}\sqrt{n}(\tilde{\theta}_n - \theta_0) - \sqrt{n}(\Psi_n(\tilde{\theta}_n) - \Psi_n(\theta_0)) - \sqrt{n}\Psi_n(\theta_0).$$

By condition (†), the middle term equals $\dot{\Psi}_0\sqrt{n}(\tilde{\theta}_n - \theta_0) + o_P(1)$ (since $\tilde{\theta}_n$ is $\sqrt{n}$-consistent, $\sqrt{n}\|\tilde{\theta}_n - \theta_0\| = O_P(1)$). Therefore

$$\dot{\Psi}_{n,0}\sqrt{n}(\hat{\theta}_n - \theta_0) = (\dot{\Psi}_{n,0} - \dot{\Psi}_0)\underbrace{\sqrt{n}(\tilde{\theta}_n - \theta_0)}_{O_P(1)} - \sqrt{n}\Psi_n(\theta_0) + o_P(1).$$

Since $\dot{\Psi}_{n,0} - \dot{\Psi}_0 \xrightarrow{p} 0$ and $\sqrt{n}(\tilde{\theta}_n - \theta_0) = O_P(1)$, the first term is $o_P(1)$. Thus

$$\dot{\Psi}_{n,0}\sqrt{n}(\hat{\theta}_n - \theta_0) = -\sqrt{n}\Psi_n(\theta_0) + o_P(1).$$

Multiplying by $\dot{\Psi}_{n,0}^{-1} \xrightarrow{p} \dot{\Psi}_0^{-1}$ and applying Slutsky's lemma gives the result. □

Remark. The one-step estimator achieves the same asymptotic distribution as the Z-estimator, regardless of the quality of the initial estimator $\tilde{\theta}_n$, as long as it is $\sqrt{n}$-consistent. This separates the problem of *consistency* (achieved by $\tilde{\theta}_n$, e.g., via method of moments) from the problem of *asymptotic efficiency* (achieved by the one-step correction).

8.3.3 Method-of-Moments Estimators

Definition (Method-of-moments estimator). Let $f = (f_1, \ldots, f_k)$ be a vector of functions, and define $e : \Theta \to \mathbb{R}^k$ by $e(\theta) = P_\theta f$. The **method-of-moments (MoM) estimator** $\hat{\theta}_n$ solves

$$\hat{P}_n f = \frac{1}{n}\sum_{i=1}^n f(X_i) = P_{\hat{\theta}_n} f = e(\hat{\theta}_n).$$

The simplest case is $f_j(x) = x^j$ (sample moments equal population moments).

Proposition (Consistency and asymptotic normality of MoM). If e is one-to-one and continuous and the LLN applies to f under P_{θ_0}, then

$$\hat{\theta}_n = e^{-1}(\hat{P}_n f) \xrightarrow{p} e^{-1}(P_{\theta_0} f) = \theta_0.$$

If additionally e^{-1} is differentiable and $\hat{P}_n f$ is asymptotically normal, the delta method gives asymptotic normality of $\sqrt{n}(\hat{\theta}_n - \theta_0)$.

Remark. MoM estimators are generally consistent under weak conditions but may have higher asymptotic variance than the MLE. They are nonetheless useful as initial values $\tilde{\theta}_n$ for one-step estimators or Newton–Raphson iterations toward the MLE.

8.4 Asymptotic Normality of Z-estimators with Nuisance Parameters

8.4.1 Setup

In many problems, the parameter of interest is $\theta \in \mathbb{R}^k$ but estimation requires a nuisance parameter η in some metric space. We have estimating equations $\Psi_n(\theta, \eta) = \hat{P}_n \psi_{\theta,\eta}$, with $\hat{\theta}_n$ solving $\hat{P}_n \psi_{\hat{\theta}_n, \hat{\eta}_n} = o_P(n^{-1/2})$ for some consistent estimator $\hat{\eta}_n$.

Theorem (Asymptotic normality with nuisance parameters). Let $\theta \in \mathbb{R}^k$ and η in a metric space. Suppose the class $\{\psi_{\theta,\eta} : \|\theta - \theta_0\| < \delta,\ d(\eta, \eta_0) < \delta\}$ is sufficiently regular (Donsker), $P\psi_{\theta_0,\eta_0} = 0$, and $\theta \mapsto P\psi_{\theta,\eta}$ is differentiable at θ_0 uniformly in η near η_0, with invertible derivative matrices $V_{\theta_0,\eta} \to V_{\theta_0,\eta_0}$.
If $\sqrt{n}\hat{P}_n \psi_{\hat{\theta}_n, \hat{\eta}_n} = o_P(1)$ and $(\hat{\theta}_n, \hat{\eta}_n) \xrightarrow{p} (\theta_0, \eta_0)$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V_{\theta_0,\eta_0}^{-1} \sqrt{n} P\psi_{\theta_0,\hat{\eta}_n} - V_{\theta_0,\eta_0}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_{\theta_0,\eta_0}(X_i) + o_P(1 + \sqrt{n}\|P\psi_{\theta_0,\hat{\eta}_n}\|).$$

Proof. The regularity assumption gives

$$\sqrt{n}(\hat{P}_n \psi_{\hat{\theta}_n,\hat{\eta}_n} - P\psi_{\hat{\theta}_n,\hat{\eta}_n}) - \sqrt{n}(\hat{P}_n \psi_{\theta_0,\eta_0} - P\psi_{\theta_0,\eta_0}) \xrightarrow{p} 0. \tag{$*$}$$

Examine the first term. Since $\sqrt{n}\hat{P}_n \psi_{\hat{\theta}_n,\hat{\eta}_n} = o_P(1)$ and $P\psi_{\theta_0,\eta_0} = 0$:

$$\sqrt{n}(\hat{P}_n \psi_{\hat{\theta}_n,\hat{\eta}_n} - P\psi_{\hat{\theta}_n,\hat{\eta}_n}) = \sqrt{n}(P\psi_{\theta_0,\hat{\eta}_n} - P\psi_{\hat{\theta}_n,\hat{\eta}_n}) - \sqrt{n}P\psi_{\theta_0,\hat{\eta}_n} + o_P(1).$$

Differentiability of $\theta \mapsto P\psi_{\theta,\eta}$ uniformly in η gives

$$= \sqrt{n}V_{\theta_0,\hat{\eta}_n}(\theta_0 - \hat{\theta}_n) - \sqrt{n}P\psi_{\theta_0,\hat{\eta}_n} + o_P(1 + \sqrt{n}\|\theta_0 - \hat{\theta}_n\|).$$

Combining with $(*)$ and using $V_{\theta_0,\hat{\eta}_n} \xrightarrow{p} V_{\theta_0,\eta_0}$:

$$\sqrt{n}(\hat{P}_n \psi_{\theta_0,\eta_0} - P\psi_{\theta_0,\eta_0}) + o_P(1) = \sqrt{n}V_{\theta_0,\eta_0}(\theta_0 - \hat{\theta}_n) - \sqrt{n}P\psi_{\theta_0,\hat{\eta}_n} + o_P(1 + \sqrt{n}\|\theta_0 - \hat{\theta}_n\|).$$

One can show $\sqrt{n}\|\theta_0 - \hat{\theta}_n\| \le O_P(1) + \sqrt{n}\|P\psi_{\theta_0,\hat{\eta}_n}\|$. Multiplying through by V_{θ_0,η_0}^{-1} gives the stated expansion. □

8.4.2 The Cost of Estimating the Nuisance Parameter

In the "nice" case where $\sqrt{n}\|P\psi_{\theta_0,\hat{\eta}_n}\| = O_P(1)$, the nuisance parameter contributes an additional variance term. Specifically, if $\sqrt{n}(\hat{\eta}_n - \eta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\eta_0})$ and $\eta \mapsto P\psi_{\theta_0,\eta}$ is differentiable at η_0 with derivative W_{θ_0,η_0}, then

$$\sqrt{n}P\psi_{\theta_0,\hat{\eta}_n} \approx W_{\theta_0,\eta_0}\sqrt{n}(\hat{\eta}_n - \eta_0),$$

and the limiting distribution of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ is

$$\begin{aligned}\sqrt{n}(\hat{\theta}_n - \theta_0) &\xrightarrow{d} \mathcal{N}(0,\ \Sigma_\theta)\,,\\ \Sigma_\theta &= V_{\theta_0,\eta_0}^{-1}\big(W_{\theta_0,\eta_0}\Sigma_{\eta_0}W_{\theta_0,\eta_0}^\top + P(\psi_{\theta_0,\eta_0}\psi_{\theta_0,\eta_0}^\top)\big)(V_{\theta_0,\eta_0}^{-1})^\top.\end{aligned}$$

The first term inside the parentheses is the additional variance from estimating $\hat{\eta}_n$.

8.4.3 Orthogonality: Eliminating the Nuisance Variance

Definition (Orthogonal estimating equations). The estimating equations $\psi_{\theta,\eta}$ are **orthogonal** to the nuisance parameter at (θ_0, η_0) if

$$\frac{\partial}{\partial\eta}P\psi_{\theta_0,\eta}\bigg|_{\eta=\eta_0} = W_{\theta_0,\eta_0} = 0.$$

When orthogonality holds, the additional variance term $V^{-1}W\Sigma_\eta W^\top(V^{-1})^\top$ vanishes, and $\sqrt{n}(\hat{\theta}_n - \theta_0)$ has the same limiting distribution as if η_0 were known:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\big(0,\ V_{\theta_0,\eta_0}^{-1}P(\psi_{\theta_0,\eta_0}\psi_{\theta_0,\eta_0}^\top)(V_{\theta_0,\eta_0}^{-1})^\top\big)\,.$$

Example (Symmetric location with scale nuisance). Suppose X_i has a density symmetric about θ_0, and we estimate θ via

$$\frac{1}{n}\sum_{i=1}^n \psi\left(\frac{X_i - \theta}{\hat{\eta}_n}\right) = 0,$$

where ψ is odd and $\hat{\eta}_n$ is a $\sqrt{n}$-consistent estimator of scale η_0. Then $W_{\theta_0,\eta_0} = \dfrac{\partial}{\partial\eta}P\psi((X - \theta_0)/\eta)\big|_{\eta_0} = 0$ by symmetry of the density and oddness of ψ, so there is no efficiency loss from estimating η.

8.4.4 More on Nuisance Parameters and Orthogonality

8.4.5 Conditional Expectation and Orthogonality

Recall the key property: for any measurable $f(X)$ and random variables X, Y with finite expectations,

$$E[f(X)(Y - E[Y \mid X])] = 0.$$

Equivalently, $E[Y \mid X]$ is the projection of Y onto the space of functions of X, and the residual $Y - E[Y \mid X]$ is orthogonal to all functions of X. Any regression can thus be written as $Y = m(X) + \varepsilon$ with $E[\varepsilon \mid X] = 0$.

8.4.6 The Partially Linear Model

Definition (Partially linear model (PLM)).

$$Y = \theta_0 X + g_{\eta_0}(W) + \varepsilon, \qquad E[\varepsilon \mid X, W] = 0.$$

The parameter of interest is $\theta_0 \in \mathbb{R}$; the nuisance function g_{η_0} is estimated alongside.

Proposition (Standard Z-estimator for the PLM). The function

$$\psi_{\theta,\eta}(w, x, y) = \begin{pmatrix} x(y - \theta x - g_\eta(w)) \\ w(y - \theta x - g_\eta(w)) \end{pmatrix}$$

defines valid estimating equations: $P\psi_{\theta_0,\eta_0} = (0, 0)^\top$.

Proof. $E[x(Y - \theta_0 X - g_{\eta_0}(W))] = E[X\varepsilon] = E[E[X\varepsilon \mid X, W]] = E[XE[\varepsilon \mid X, W]] = 0$. Similarly for the W component. □

Under $E[\varepsilon^2 \mid X, W] = \sigma^2$, the asymptotic variance of $\sqrt{n}(\hat{\theta}_n - \theta_0)$ (the $(1, 1)$ entry of the full variance matrix) from Theorem 22.1 is

$$\sigma^2_{\theta_0}(\eta_0) = \sigma^2 \cdot \frac{E[\dot{g}_{\eta_0}(W)W]^2 E[X^2] + E[\dot{g}_{\eta_0}(W)X]^2 E[W^2] - 2E[XW]E[\dot{g}_{\eta_0}(W)W]E[\dot{g}_{\eta_0}(W)X]}{\left(E[X^2]E[\dot{g}_{\eta_0}(W)W] - E[XW]E[\dot{g}_{\eta_0}(W)X]\right)^2}.$$

This is complicated and depends on the derivative of the nuisance function.

8.4.7 Orthogonal Estimation of the PLM

Definition (Robinson's orthogonal estimating equations). Define $h_{\gamma_0}(W) = E[Y \mid W]$ and $m_{\beta_0}(W) = E[X \mid W]$. Consider

$$\psi_{\theta,\gamma,\beta}(w, x, y) = (y - h_\gamma(w) - \theta(x - m_\beta(w)))(x - m_\beta(w)).$$

The motivation comes from the PLM identity:

$$E[Y \mid W] = \theta_0 E[X \mid W] + g_{\eta_0}(W),$$

so $Y - h_{\gamma_0}(W) = \theta_0(X - m_{\beta_0}(W)) + \varepsilon$. Defining residuals $\tilde{Y} = Y - h_{\gamma_0}(W)$ and $\tilde{X} = X - m_{\beta_0}(W)$, this becomes $\tilde{Y} = \theta_0 \tilde{X} + \varepsilon$, and the estimating equation $\psi_\theta(\tilde{x}, \tilde{y}) = \tilde{x}(\tilde{y} - \theta\tilde{x})$ is simply least squares on the residuals.

Proposition (Validity and orthogonality).

1. $P\psi_{\theta_0,\gamma_0,\beta_0} = 0$.
2. **Orthogonality:** $\dfrac{\partial}{\partial\beta} P\psi_{\theta_0,\gamma_0,\beta_0} = \dfrac{\partial}{\partial\gamma} P\psi_{\theta_0,\gamma_0,\beta_0} = 0$.
3. $\dfrac{\partial}{\partial\theta} P\psi_{\theta_0,\gamma_0,\beta_0} = -E[(X - m_{\beta_0}(W))^2]$.
4. $P(\psi^2_{\theta_0,\gamma_0,\beta_0}) = E[\varepsilon^2(X - m_{\beta_0}(W))^2]$.

Proof. (1) $P\psi_{\theta_0,\gamma_0,\beta_0} = E[(Y - h_{\gamma_0}(W) - \theta_0(X - m_{\beta_0}(W)))(X - m_{\beta_0}(W))] = E[\varepsilon\tilde{X}] = 0$ by $E[\varepsilon \mid X, W] = 0$ and $\tilde{X}$ being a function of X, W.
(2) At $(\theta_0, \gamma_0, \beta_0)$, the residual $y - h_\gamma(w) - \theta(x - m_\beta(w))$ equals ε regardless of θ_0. Differentiating $P\psi$ with respect to β at β_0 involves $\dfrac{\partial}{\partial\beta}E[\varepsilon\tilde{X}]$, which evaluates to $E[\varepsilon(-\dot{m}_{\beta_0}(W))] = 0$ since $E[\varepsilon \mid W] = 0$. Similarly for γ.
(3) $\dfrac{\partial}{\partial\theta}P\psi_{\theta_0,\gamma_0,\beta_0} = -E[(X - m_{\beta_0}(W))^2]$.
(4) $P(\psi^2_{\theta_0,\gamma_0,\beta_0}) = E[\varepsilon^2(X - m_{\beta_0}(W))^2]$. □

Theorem (Asymptotic normality of the orthogonal PLM estimator). Suppose $\sqrt{n}((\hat{\gamma}_n, \hat{\beta}_n) - (\gamma_0, \beta_0)) = O_P(1)$. By Proposition 8.4.7 and orthogonality (no variance contribution from estimating γ or β):

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{E[\varepsilon^2(X - m_{\beta_0}(W))^2]}{E[(X - m_{\beta_0}(W))^2]^2}\right).$$

If additionally $E[\varepsilon^2 \mid X, W] = \sigma^2$, this simplifies to

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma^2}{E[(X - m_{\beta_0}(W))^2]}\right).$$

Proof. By Theorem 22.1, orthogonality ($W_{\theta_0,\gamma_0,\beta_0} = 0$) eliminates the nuisance variance term. The asymptotic variance is therefore $V^{-1}P(\psi^2)(V^{-1})^\top$ with $V = -E[\tilde{X}^2]$:

$$\frac{E[\varepsilon^2\tilde{X}^2]}{E[\tilde{X}^2]^2} = \frac{\sigma^2 E[\tilde{X}^2]}{E[\tilde{X}^2]^2} = \frac{\sigma^2}{E[\tilde{X}^2]}.$$

□

Remark. This is Robinson's (1988) estimator for the PLM. The variance $\sigma^2/E[\tilde{X}^2]$ is the same as if $m_{\beta_0}(W)$ were known — orthogonality gives *rate-free* robustness to the nuisance estimation, meaning $\hat{\gamma}_n$ and $\hat{\beta}_n$ need only be $\sqrt{n}$-consistent (e.g., nonparametric estimators at rates slower than $n^{-1/2}$ can still be used in practice).

Chapter 9

Special Topics in Likelihood

9.1 Time-to-Event Data and Survival Analysis

9.1.1 Survival and Hazard Functions

Let $T_1, \ldots, T_n \geq 0$ be event times with unknown CDF F and PMF/PDF f.

Definition (Survival function).

$$S(t) = P(T > t) = \begin{cases} \displaystyle\int_t^\infty f(s)\,\mathrm{d}s & T \text{ continuous}, \\ \displaystyle\sum_{t_k \geq t} f_{t_k} & T \text{ discrete}. \end{cases}$$

Definition (Hazard rate).

$$h(t) = \frac{f(t)}{P(T \geq t)}.$$

This is the instantaneous rate of event occurrence given survival to time t.

Proposition (Survival from hazard). In the discrete case, for $t \in [t_j, t_{j+1})$:

$$S(t) = \prod_{k: t_k \leq t} (1 - h(t_k)), \qquad \frac{S(t_j)}{S(t_{j-1})} = 1 - h(t_j).$$

In the continuous case:

$$S(t) = \exp\left(-\int_0^t h(s)\,\mathrm{d}s\right), \qquad \frac{S(t)}{S(u)} = \exp\left(-\int_u^t h(s)\,\mathrm{d}s\right), \quad u < t.$$

Proof of continuous case. Partition $[0, t]$ into N sub-intervals of length $\mathrm{d}t_N$ centered at t_k. Approximating $S_N(t) = \prod_{k: t_k \leq t} (1 - h(t_k)\mathrm{d}t_N)$ and taking logs:

$$\log S_N(t) = \sum_{k: t_k \leq t} \log(1 - h(t_k)\mathrm{d}t_N) = -\sum_{k: t_k \leq t} h(t_k)\mathrm{d}t_N + O(\mathrm{d}t_N^2) \to -\int_0^t h(s)\,\mathrm{d}s$$

as $N \to \infty$. $\square$

9.1.2 Censored Data and the Kaplan–Meier Estimator

Definition (Right censoring). Data are **right-censored** when the event of interest is not observed for some subjects. Observations are recorded as pairs $Z_i = (T_i, C_i)$ where

$$C_i = \begin{cases} 0 & \text{event observed,} \\ 1 & \text{censored (event time } T_i > t_i^*). \end{cases}$$

Definition (Risk set and likelihood). Sort observed times as $t_{(1)} \leq \cdots \leq t_{(n)}$. Let $\mathcal{R}_j = \{i : T_i \geq t_{(j)}\}$ be the **risk set** at time $t_{(j)}$, and $d_j = |\{i : T_i = t_{(j)}, C_i = 0\}|$ the number of events.
The likelihood of $(z_1, \ldots, z_n)$ factors as

$$L_n(F) = \prod_{j:\text{distinct } t_{(j)}} h(t_{(j)})^{d_j}(1 - h(t_{(j)}))^{|\mathcal{R}_j| - d_j}.$$

Proof. The likelihood for observation i is $f(t_i)^{1-c_i}S(t_i)^{c_i}$. Conditioning on the observed event times and allowing probability mass only at uncensored times, substituting $f(t_{(j)}) = h(t_{(j)})\prod_{k<j}(1 - h(t_{(k)}))$ and collecting terms over the risk sets gives the product form. □

Theorem (Kaplan–Meier estimator). The MLE of the hazard rate is $\hat{h}(t_{(j)}) = d_j/|\mathcal{R}_j|$. The resulting **Kaplan–Meier (KM)** estimator of the survival function is

$$\hat{S}(t) = \prod_{j:t_{(j)} \leq t} \left(1 - \frac{d_j}{|\mathcal{R}_j|}\right), \quad t_{(j)} \leq t < t_{(j+1)}.$$

Proof. The likelihood $L_n(F)$ factors over distinct event times as a product of binomial-type terms. Maximizing each factor $h(t_{(j)})^{d_j}(1 - h(t_{(j)}))^{|\mathcal{R}_j| - d_j}$ separately over $h(t_{(j)}) \in [0, 1]$ gives $\hat{h}(t_{(j)}) = d_j/|\mathcal{R}_j|$. Substituting into the discrete survival formula gives the KM estimator. □

Proposition (KM = empirical CDF under no censoring). If there is no censoring, the KM estimator equals $1 - \hat{F}_n(t)$.

Proof. With no censoring, $|\mathcal{R}_j| = n - \overline{d}_{j-1}$ where $\overline{d}_j = \sum_{i=1}^{j} d_i$. Then

$$\hat{S}(t) = \prod_{j:t_{(j)} \leq t} \frac{n - \overline{d}_j}{n - \overline{d}_{j-1}} = \frac{n - \overline{d}_j}{n} = 1 - \frac{\overline{d}_j}{n} = 1 - \hat{F}_n(t),$$

using telescoping. □

9.1.3 Proportional Hazards Model

Definition (Cox proportional hazards model). With covariates $x_i \in \mathbb{R}^p$, assume individual i has hazard

$$h_i(t) = h_0(t)\, e^{x_i\beta},$$

where $h_0(t)$ is an unspecified baseline hazard. The **hazard ratio** of individuals i and j is constant: $h_i(t)/h_j(t) = e^{(x_i - x_j)\beta}$.

Definition (Partial likelihood). Conditioning on the risk sets and assuming no ties, the **partial likelihood** is

$$L_n(\beta) = \prod_{j=1}^{n} \frac{e^{x_{(j)}\beta}}{\sum_{i\in\mathcal{R}_j} e^{x_i\beta}},$$

where $x_{(j)}$ is the covariate vector of the individual with event at time $t_{(j)}$.

Remark. The baseline hazard $h_0(t)$ cancels from the partial likelihood, so β can be estimated without estimating h_0. This makes the Cox model semiparametric: parametric in β, nonparametric in h_0. Standard asymptotic normality arguments apply to the partial MLE $\hat{\beta}$.

9.2 Missing Data, Latent Variables, and the EM Algorithm

9.2.1 Setup

Suppose $X^n = (X_1, \ldots, X_n)$ are observed and $Z^m = (Z_1, \ldots, Z_m)$ are unobserved (latent or missing). The likelihood of x^n is

$$L_n(\theta) = f(x^n; \theta) = \int f(x^n, z^m; \theta)\, \mathrm{d}z^m.$$

9.2.2 Gaussian Mixture Models

Definition (Gaussian mixture model). Each X_i is drawn i.i.d. from a K-component mixture:

$$f(x_i; \theta) = \sum_{k=1}^{K} \pi_k\, \varphi(x_i; \mu_k, \Sigma_k),$$

where $\varphi(\cdot; \mu, \Sigma)$ is the Gaussian PDF, $\pi_k \in (0, 1)$, and $\sum_k \pi_k = 1$.

Definition (Latent variable representation). Each observation has a latent **component assignment** $Z_i = (Z_{i1}, \ldots, Z_{iK})$ from the categorical distribution with probabilities (π_k), where $Z_{ik} \in \{0, 1\}$ and $\sum_k Z_{ik} = 1$. The complete-data likelihood is

$$f(x^n, z^n; \theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[\varphi(x_i; \mu_k, \Sigma_k)\, \pi_k\right]^{z_{ik}}.$$

9.2.3 The EM Algorithm

Definition (EM algorithm). Initialize $\theta^{(0)}$. For $j = 1, 2, \ldots$ until convergence:

1. **E-step:** Compute
$$J(\theta \mid \theta^{(j-1)}) = E_{\theta^{(j-1)}}\left[\log \frac{f(X^n, Z^m; \theta)}{f(X^n, Z^m; \theta^{(j-1)})} \;\middle|\; X^n = x^n\right].$$
2. **M-step:** Set $\theta^{(j)} = \arg\max_{\theta} J(\theta \mid \theta^{(j-1)})$.

Proposition (E-step for Gaussian mixtures). For the Gaussian mixture model, the E-step reduces to computing the **responsibilities**:
$$\gamma_{ik}(\theta) := E_\theta[Z_{ik} \mid X^n = x^n] = E_\theta[Z_{ik} \mid X_i = x_i] = \frac{\pi_k\, \varphi(x_i; \mu_k, \sigma_k^2)}{\sum_{\ell=1}^K \pi_\ell\, \varphi(x_i; \mu_\ell, \sigma_\ell^2)}.$$

Proof. By Bayes' rule, $E_\theta[Z_{ik} \mid X_i = x_i] = P_\theta(Z_{ik} = 1 \mid X_i = x_i)$. Since $P(Z_{ik} = 1) = \pi_k$ and $f(x_i \mid Z_{ik} = 1; \theta) = \varphi(x_i; \mu_k, \sigma_k^2)$, Bayes' theorem gives the expression above. □

Proposition (M-step for Gaussian mixtures (estimating means)). When only μ_k are to be estimated (with π_k, σ_k^2 known), the M-step gives
$$\mu_k^{(j+1)} = \frac{\sum_{i=1}^n x_i\, \gamma_{ik}(\theta^{(j)})}{\sum_{i=1}^n \gamma_{ik}(\theta^{(j)})}.$$

Proof. The joint log-likelihood contains, for the terms involving μ_k:
$$E_{\theta^{(j)}}[\log f(X^n, Z^n; \theta) \mid X^n = x^n] \ni -\sum_{i=1}^n \sum_{k=1}^K \frac{\gamma_{ik}(\theta^{(j)})}{2\sigma_k^2}(x_i - \mu_k)^2.$$
Differentiating with respect to μ_k and setting to zero gives the weighted mean formula. □

9.2.4 Monotone Ascent of EM

Theorem (EM increases the likelihood). At each step j, $\log f(x^n; \theta^{(j+1)}) \geq \log f(x^n; \theta^{(j)})$. Equivalently,
$$\log \frac{f(x^n; \theta^{(j+1)})}{f(x^n; \theta^{(j)})} = J(\theta^{(j+1)} \mid \theta^{(j)}) + D\big(f(z^m \mid x^n; \theta^{(j)}) \parallel f(z^m \mid x^n; \theta^{(j+1)})\big) \geq 0,$$
where $D(p\|q) = \int p(z) \log(p(z)/q(z))\, \mathrm{d}z \geq 0$ is the KL divergence.

Proof. Write $\log f(x^n; \theta) = \log f(x^n, z^m; \theta) - \log f(z^m \mid x^n; \theta)$. Taking $E_{\theta^{(j)}}[\,\cdot \mid X^n = x^n]$ of both sides at $\theta = \theta^{(j+1)}$ and $\theta = \theta^{(j)}$, then taking the difference:
$$\log \frac{f(x^n; \theta^{(j+1)})}{f(x^n; \theta^{(j)})} = J(\theta^{(j+1)} \mid \theta^{(j)}) - J(\theta^{(j)} \mid \theta^{(j)}) + D\big(f(z^m \mid x^n; \theta^{(j)}) \parallel f(z^m \mid x^n; \theta^{(j+1)})\big).$$
Since $\theta^{(j+1)}$ maximizes $J(\cdot \mid \theta^{(j)})$, $J(\theta^{(j+1)} \mid \theta^{(j)}) \geq J(\theta^{(j)} \mid \theta^{(j)}) = 0$. Non-negativity of KL divergence completes the proof. □

Remark. If the log-likelihood is concave, EM converges to the global maximum. Otherwise it converges to a stationary point (local max, local min, or saddle point) depending on initialization.